

Hint-AUC: Deterministic Region-based Hint Generation for Line Art Colorization Evaluation

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This supplementary material provides additional details not included in the main paper due to space constraints. First, we present the full derivation of the proposed deterministic hint generation and additional analyses of Hint-AUC, including the sensitivity to the α sampling grid, robustness to the region segmenter, and seed sensitivity under stochastic hint sampling. We then provide the architectural details of DetFill and additional details of the formative study. Finally, we include extended quantitative results on natural images, more qualitative results from our experiments, and additional training comparisons. The code will be made publicly available once the paper is accepted.

I. DETAILS OF DETERMINISTIC HINT GENERATION

This section describes the hint-generation pipeline and the stored inputs used for evaluation; the archived hint and label maps define the reference inputs for the reported results. For region segmentation G_r , we use the Felzenszwalb algorithm [1] with scale = 100, $\sigma = 0.5$, and a minimum region size of 100 pixels. The resulting regions are ordered deterministically by descending pixel area, which fixes the sequence in which hints are added. For each region we extract a one-pixel-wide skeleton using Zhang–Suen thinning [2] (the `skeletonize` operator), thicken it with a 3×3 dilation (one iteration) so that the longest-path extraction operates on a stable input, and then extract the longest path along the skeleton with FilFinder2D [3] (pruned by length, with a branch and skeleton threshold of 3 pixels; the returned medial-axis path is one pixel wide). The scribble hint is this longest path intersected with the region mask. The dot hint is a single pixel at the mean row and column coordinates of the longest path, truncated to integers. The hint color is the mean RGB color, in the ground-truth image, of the region containing the hint pixel.

a) Integer conversion, tie-breaking, and pixel mapping:

At ratio α , we select $k = \lfloor \alpha n \rfloor$ of the n region labels in the 64×64 hint map (Algorithm 1). For $\alpha > 0$, images with $n < 1/\alpha$ receive no hints. This occurs for one of the 3,000 test images at $\alpha = 1\%$.

Regions are enumerated in ascending label order and sorted by decreasing area with the default sorting routine of the array library (NumPy). Nearly every test image has an area tie at a selection boundary for at least one grid ratio. The order of tied regions depends on the library version: versions 1.26 and 2.0 of the array library select different regions in most images at intermediate ratios. The affected regions have median areas of 2–21 pixels at 64×64 , depending on α . The mean swapped area is 0.1–0.8% of the hinted area, with a maximum of

14% for one image at $\alpha = 1\%$. We did not evaluate the effect on colorization scores. The DetFill scribble results were produced with version 2.0.2 of the array library, which the release environment pins. Each method’s data loader applies this selection in its own software environment, so the tied smallest regions may differ between methods at intermediate ratios by the same small margin; we did not verify the per-ratio selections across methods.

For equal-length candidates, the longest-path extraction (FilFinder2D, version 1.7.2) returns the first longest path found during graph traversal. The medial-axis extraction inside this step, however, breaks ties between pixels in a randomized order, and region labels were assigned from a shuffled palette. Neither step was seeded. Regenerating hints from the source images is therefore not bit-exact. The stored hint and label maps define the reference inputs for all reported results.

We place each dot at the mean row and column coordinates of the longest-path pixels before clipping the path to its region. Both coordinates are truncated to integers without projection onto the region. A dot can therefore fall outside its source region. The dot is selected together with the region that contains the dot pixel and takes the mean color of that region. All models use the same stored hint maps. As a proxy for out-of-region placement, we recomputed truncated centroids from the stored, region-clipped scribble masks. The centroids fell outside their source regions for 3.0% of these masks. Because this calculation uses clipped scribbles rather than the original paths, it does not directly measure the out-of-region rate of the stored dots. Skeleton extraction failed for 0.26% of regions, most of which had at most four pixels; these regions produced no stroke or dot.

During training, we independently draw $u \sim \mathcal{U}[0,1)$ for each example, randomly permute the n labels, and retain the first $p = \lfloor nu \rfloor$ (main paper, Eq. (6)). Thus, p is uniform on $\{0, \dots, n-1\}$. Training includes zero-hint inputs but excludes the full-hint case, which is used only at evaluation.

b) *Archived implementation:* The release includes the training, validation, and test image lists; DetFill configurations for both hint types and image domains; the reported scribble and dot checkpoints; hint-generation scripts; and an environment file with pinned PyTorch, NumPy, scikit-image, OpenCV, and FilFinder versions. FilFinder is fixed at version 1.7.2. DetFill uses inference seed 1234. We use the official baseline inference scripts: PaintsTorch is deterministic, Diffusart calls `torch.manual_seed(2)` before generating the initial noise, and ColorizeDiffusion v1/v2 use unseeded noise. Each model is evaluated once per image, hint ratio, and sketch source; see Sec. VI-A of the main paper.

TABLE I

SENSITIVITY OF HINT-AUC TO THE α SAMPLING GRID ON DANBOORU2021 USING THE DETFILL CHECKPOINT THAT IS ALSO USED FOR THE SCRIBBLE RESULTS IN TABLE II OF THE MAIN PAPER.

α grid	N	MSE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	OpenCLIP $\uparrow$	DINO $\uparrow$	DreamSim $\downarrow$
Uniform, $\Delta\alpha=0.02$ (dense ref.)	51	0.0148 ± 0.0049	19.779 ± 1.420	0.7254 ± 0.0858	0.1807 ± 0.0466	0.9549 ± 0.0099	0.9514 ± 0.0112	0.0823 ± 0.0207
Front-loaded [†]	8	0.0149 ± 0.0050	19.723 ± 1.414	0.7235 ± 0.0860	0.1829 ± 0.0472	0.9543 ± 0.0101	0.9509 ± 0.0114	0.0839 ± 0.0212
Uniform, $\Delta\alpha=0.04$	26	0.0154 ± 0.0050	19.745 ± 1.416	0.7246 ± 0.0857	0.1817 ± 0.0465	0.9546 ± 0.0099	0.9512 ± 0.0112	0.0833 ± 0.0206
Uniform-like, nominal $\Delta\alpha=0.08$	14	0.0168 ± 0.0051	19.653 ± 1.408	0.7226 ± 0.0856	0.1844 ± 0.0461	0.9537 ± 0.0098	0.9506 ± 0.0112	0.0864 ± 0.0204
Uniform-like, nominal $\Delta\alpha=0.16$	8	0.0199 ± 0.0055	19.410 ± 1.389	0.7174 ± 0.0853	0.1915 ± 0.0456	0.9513 ± 0.0098	0.9489 ± 0.0112	0.0945 ± 0.0201

[†] Front-loaded grid: $\alpha \in \{0.00, 0.01, 0.03, 0.05, 0.10, 0.25, 0.50, 1.00\}$. All rows are evaluated using the same DetFill checkpoint with 96 base channels; the scribble results in Table II of the main paper are computed with this same checkpoint, so the front-loaded row coincides with the DetFill scribble scores reported there. Thus, the row-wise differences in this table isolate the effect of the α sampling grid rather than checkpoint variation. Values are reported as mean $\pm$ sample standard deviation over the three sketch sources: XDoG, sketch simplification, and SketchKeras, with $n = 3$ and divisor $n - 1$. The dense uniform grid with $\Delta\alpha = 0.02$ is used as the numerical reference for the discussion. For uniform-like grids whose nominal step size does not exactly divide 1, the endpoint $\alpha = 1.00$ is included and the final interval may be shorter.

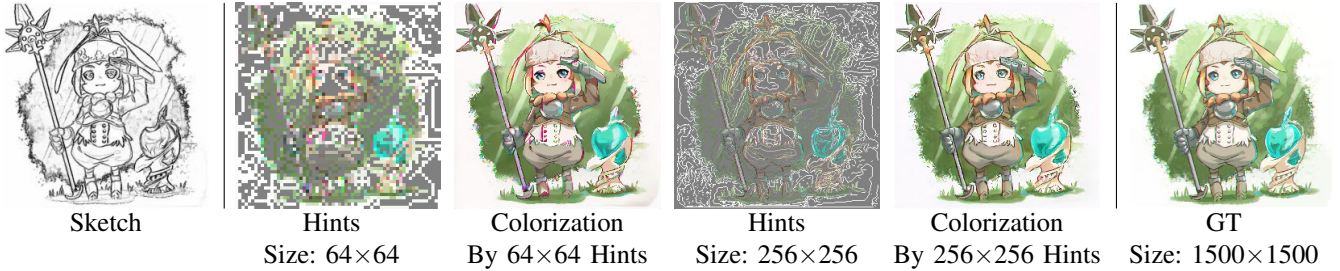


Fig. 1. **Enhancing colorization by changing the resolution of the hint:** The granularity of the hints depends on the hint resolution, with higher-resolution hints leading to improved colorization results.

TABLE II

SEGMENTATION ANALYSIS FOR DETERMINISTIC HINT GENERATION. WE SUMMARIZE EACH ALGORITHM'S STATISTICAL DATA USING A SUBSET OF DANBOORU2021 [4]

	Avg. Scribble Counts	Avg. Difference of Color	Avg. Scribble Length
SLIC [5]	185.80 ± 23.33	0.48 ± 0.09	8.20 ± 4.56
Quickshift [6]	2439.33 ± 945.87	0.15 ± 0.10	1.82 ± 1.36
Watershed [7]	244.88 ± 7.24	0.47 ± 0.08	6.48 ± 2.86
Felzenszwalb [1]	744.15 ± 452.25	0.26 ± 0.09	3.70 ± 7.31

II. ANALYSIS

A. Sensitivity of Hint-AUC to the α Sampling Grid

We further examine whether Hint-AUC is sensitive to the discrete sampling grid of the hint ratio α . To keep this grid-sensitivity analysis controlled, we use one fixed DetFill checkpoint with the same architecture and 96 base channels as the scribble model used in the main paper. This checkpoint is the model used for the scribble results in Table II of the main paper; accordingly, the front-loaded row in Table I coincides with the DetFill scribble scores reported there. Since every row in Table I uses the same checkpoint, dataset, sketch sources, and evaluation metrics, row-wise differences isolate the effect of the α sampling grid.

Table I reports Hint-AUC values computed using several alternative α grids. We use the dense uniform grid with $\Delta\alpha = 0.02$ as a numerical reference. The front-loaded grid used in the main evaluation closely matches this dense uniform reference despite using only eight sampling points. Compared with the dense reference, the front-loaded grid changes MSE from 0.0148 to 0.0149, PSNR from 19.779 to 19.723, SSIM from 0.7254 to 0.7235, LPIPS from 0.1807 to 0.1829, OpenCLIP from 0.9549 to 0.9543, DINO from 0.9514 to 0.9509,

and DreamSim from 0.0823 to 0.0839. These differences are small relative to the standard deviations over the three sketch sources, indicating that increasing the number of uniformly sampled points does not materially change the estimated Hint-AUC in this controlled setting.

In contrast, the coarsest uniform-like grid with the same number of sampling points as the front-loaded grid shows larger deviations, especially for MSE, LPIPS, and DreamSim. This behavior is consistent with the metric-hint response observed under dense sampling: the metric response changes most rapidly in the low- α region and gradually saturates as more hints are added. A coarse uniform-like grid therefore under-samples the high-curvature sparse-hint regime. The front-loaded grid allocates more samples to this region, but Hint-AUC is computed using the trapezoidal rule, where each interval contributes in proportion to its width in α . Therefore, the front-loaded grid does not assign additional AUC weight to sparse-hint settings simply because they are sampled more densely. Instead, it improves the numerical approximation in the high-curvature low- α region while preserving the area weighting over the full α range.

Algorithm 1 Deterministic Region-based Hint Generation (as implemented: all hints are built once, then selected by α)

Require: ground-truth color image $\mathbf{x}$; hint type $\tau \in \{\text{scribble}, \text{dot}\}$; hint ratio α

Ensure: color-hint map H_α , binary hint mask M_α

- 1: $\mathbf{x}_r \leftarrow \text{Felzenszwalb}(\mathbf{x}; \text{scale}=100, \sigma=0.5, \text{min_size}=100)$
 $\triangleright$ label map; hints are built on its 64×64 nearest-neighbor downsampling
- 2: $\{R_1, \dots, R_n\} \leftarrow$ all n regions of the label map
 $\triangleright$ label(q) denotes the region of pixel q
- 3: $\mathbf{H} \leftarrow \mathbf{0}$; $\mathbf{M} \leftarrow \mathbf{0}$ $\triangleright$ stage 1: build all hints once
- 4: **for** $i = 1$ to n **do**
- 5: $c_i \leftarrow \frac{1}{|R_i|} \sum_{p \in R_i} \mathbf{x}(p)$ $\triangleright$ average region color
- 6: $s_i \leftarrow \text{skeletonize}(R_i)$ $\triangleright$ Zhang–Suen thinning
- 7: $s_i \leftarrow \text{dilate}(s_i, 3 \times 3)$ $\triangleright$ thicken for path extraction
- 8: $\pi_i \leftarrow \text{FilFinder.longest_path}(s_i)$ $\triangleright$ if extraction fails or $\pi_i = \emptyset$, region i receives no hint (skip)
- 9: **if** $\tau = \text{scribble}$ **then**
- 10: $\Omega_i \leftarrow \pi_i \cap R_i$ $\triangleright$ one-pixel-wide path inside the region
- 11: **else** $\triangleright$ dot
- 12: $\Omega_i \leftarrow \{\text{meanpixel}(\pi_i)\}$ $\triangleright$ mean coordinates of π_i truncated to integers; no projection onto R_i
- 13: **end if**
- 14: $M[\Omega_i] \leftarrow 1$; $\mathbf{H}[q] \leftarrow c_{\text{label}(q)}$ for $q \in \Omega_i$ $\triangleright$ color of the region containing the hint pixel ($= c_i$ unless the dot left R_i)
- 15: **end for**
- 16: order the labels by decreasing area (ties follow the sorting routine of the implementation); $\mathcal{I}_\alpha \leftarrow$ the first $k = \lfloor \alpha n \rfloor$ labels $\triangleright$ stage 2: selection; Table III uses ascending label order instead
- 17: $A_\alpha(q) \leftarrow \mathbf{1}[\text{label}(q) \in \mathcal{I}_\alpha]$; $\mathbf{M}_\alpha \leftarrow \mathbf{M} \odot A_\alpha$; $\mathbf{H}_\alpha \leftarrow \mathbf{H} \odot A_\alpha$ $\triangleright$ hint pixels are kept by the label of the pixel they occupy
- 18: **return** $\mathbf{H}_\alpha$, $\mathbf{M}_\alpha$

B. Influence of Hint Resolution

Figure 1 visualizes the colorization results with different deterministic hint resolutions. We train two models with 64×64 sized hints and 256×256 hints, respectively. As we see in Figure 1, the 256×256 hints capture the illustration better than 64×64 hints. The colorization result with 256×256 hints is better than 64×64 hints when comparing details such as jacket, hair color, spear, and shoes. The result implies that it is important to use high resolution hints on both training the model and evaluating more minor regions.

C. Ablation Study

Figure 2 shows the results when we lower the number of base channels of DetFill to ($C = 32$). In this case, we observe noticeable blob noise in the colorized image, furthermore, when examining the cat image (Fig. 2, third row), we can see our lightweight version of DetFill struggles to properly handle the scribbles, and we can see artifacts that correspond to the input scribbles. Although these issues are mitigated when the

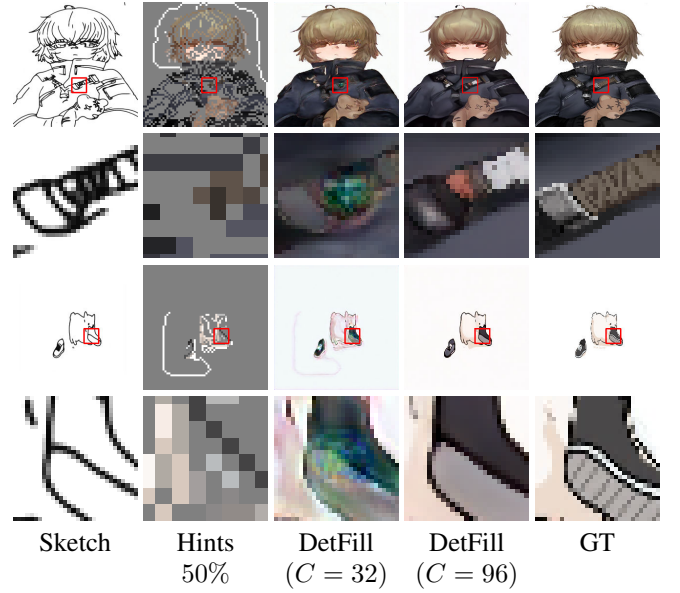


Fig. 2. **Lightweight ($C = 32$) DetFill colorization analysis:** We observed that when the DetFill’s base channel size is set to $C = 32$, DetFill produces some noise that cannot be completely denoised.

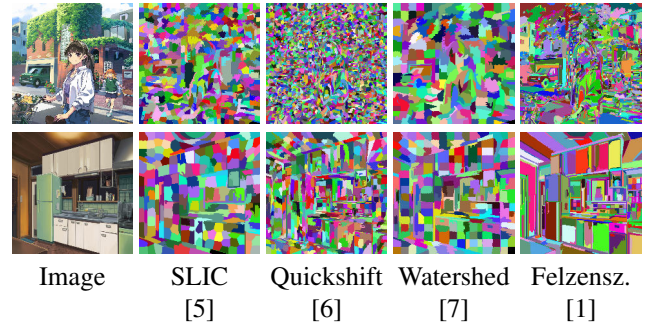


Fig. 3. **Comparison of different region segmentation approaches.** We find that the Felzenszwalb approach gives the most consistent performance.

number of base channels is increased to ($C = 96$), further improvements are necessary for more consistent colorization.

D. Segmentation Algorithm for Hint Generation

We analyzed four heuristic segmentation algorithms to identify the most suitable one for adoption. We chose these methods because they work equally well on both illustrative and real-world images, whereas deep learning-based approaches [9], [10] tend to be specialized for particular types of segmentation.

Table II summarizes three statistics: (1) the average number of scribbles, (2) the mean color difference between each scribble and its corresponding RGB value, and (3) the average length of deterministic hints around images from Danbooru2021 [4]. While Quickshift [6] achieves the lowest average color difference, it requires substantially more scribbles than the other algorithms.

Figure 3 shows the segmentation results produced by each algorithm. Notably, Felzenszwalb’s method [1] accurately separates the image into object-level regions in both examples. It is important to generate hints for both large and small

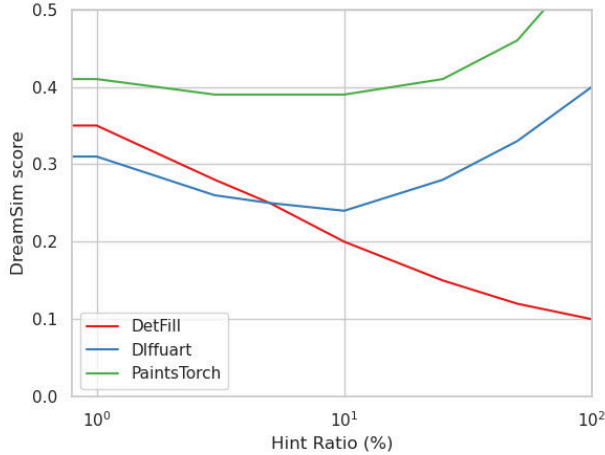


Fig. 4. **Importance of using different hint ratios at Hint-AUC:** We pick up the DreamSim [8] scores and visualize the scores with increasing hint ratios. We see that the best performing approach depends on the hint ratio. Only our approach shows monotonically increasing performance as more hints are provided.

regions. Because only Felzenszwalb’s segmentation yields regions at both scales, it is the most suitable choice to fulfill the requirement. Considering all three metrics with the visualization results, Felzenszwalb [1] offers the most balanced performance. Accordingly, we use Felzenszwalb [1] to generate our deterministic hints. Moreover, unlike illustration-specific segmenters such as DanbooRegion, Felzenszwalb is illustration-agnostic and also applies to non-illustration inputs, an additional reason we retain it as the default.

E. Importance of Different Hint Ratios

Figure 4 visualizes the DreamSim scores with dot hints under a deterministic hint sampling strategy with increasing hint ratios. We observe that the relative ranking of the approaches under DreamSim is contingent upon the hint ratio. Specifically, Diffuort scores are lower than those of DetFill when the hint ratio is under 5%. However, DetFill outperforms Diffuort when the hint ratio exceeds 5%. These findings highlight the importance of utilizing Hint-AUC for a more comprehensive comparison of colorization results, and our training approach improves performance with many hints, while other approaches tend to drop in performance.

III. ROBUSTNESS TO THE REGION SEGMENTER

The deterministic hints used by our Hint-AUC are sampled from a region segmentation of the target image, so a natural concern is whether the reported scores are an artifact of the particular segmentation backbone. To examine this, we retrain the proposed model (DetFill) under three different region segmenters—Felzenszwalb (the default segmenter of our pipeline; see Section I), DanbooRegion, and SLIC—and cross-evaluate each trained model under all three segmenters, forming a 3×3 train/evaluation matrix.

Table III reports the resulting Hint-AUC (mean $\pm$ sample standard deviation over the three sketch sources) for three

error-style metrics (MSE, LPIPS, and DreamSim; lower is better), integrated over the same eight-point hint-ratio grid $\alpha \in \{0, 0.01, 0.03, 0.05, 0.10, 0.25, 0.50, 1.00\}$ as in the main paper. Two trends emerge. First, the choice of evaluation segmenter is the dominant factor: evaluating under Felzenszwalb yields the lowest errors, whereas SLIC is consistently the most challenging setting (e.g., for the DanbooRegion-trained model the DreamSim Hint-AUC degrades from 0.094 under Felzenszwalb to 0.155 under SLIC). Second, the model is comparatively robust to the training segmenter: for a fixed evaluation segmenter, changing the segmenter used during training perturbs the Hint-AUC only mildly (e.g., evaluating under Felzenszwalb, MSE changes by less than 0.005 between the DanbooRegion- and SLIC-trained models). Taken together, this indicates that DetFill’s scores and their qualitative trends are robust across region-respecting segmenters, rather than an artifact of the specific Felzenszwalb segmenter, and that Hint-AUC tracks colorization quality consistently across region-respecting backbones. Since this analysis retrains and evaluates a single model, it characterizes segmenter sensitivity at the model level; it does not by itself establish segmenter-invariance of the full cross-model ranking, which would require retraining every baseline under each segmenter.

Figures 5 and 6 give a qualitative view for two representative test images. For each example and each of the three segmenters, we show the region map, the corresponding color hint, and the colorization produced at every hint ratio α , alongside the ground truth. Although the three segmenters partition each image into visibly different regions—and therefore place the color hints differently—the colorizations become increasingly faithful and mutually consistent as α grows, in line with the quantitative robustness in Table III.

Qualitatively, SLIC partitions the line art into approximately uniform superpixels that often do not follow semantic or color boundaries; under the size-sorted protocol this can delay broad coverage of a semantic region at small α , leaving the model only weakly conditioned there (Figures 5 and 6). This helps explain why SLIC is consistently the most challenging evaluation segmenter in Table III: the standard front-loaded grid samples this low- α regime densely and thus exposes the failure rather than averaging it away. We regard segmentation quality as a limitation of Hint-AUC rather than claiming segmenter invariance.

IV. SEED SENSITIVITY AND SELECTION-RULE COMPARISON

Table IV reports the per-ratio metric values across the front-loaded eight-point hint-ratio grid $\alpha \in \{0, 1, 3, 5, 10, 25, 50, 100\}\%$ together with the HAUC trapezoidal integral, for both the proposed size-sort hint selection and uniformly random region selection. The random row aggregates 10 independent hint-sampling seeds as mean $\pm$ sample SD (only the hint-selection seed varies; the diffusion-sampling seed and all other settings are fixed), whereas the size-sort row has no standard deviation because the proposed selection is deterministic: given the same image and hint ratio it always selects the same regions, which is

TABLE III
SEGMENTATION-DEPENDENCY CROSS-EVALUATION (COMPACT). HINT-AUC (MEAN $\pm$ SAMPLE SD OVER THE THREE SKETCH SOURCES) FOR THREE LOWER-IS-BETTER METRICS OVER THE EIGHT-POINT FRONT-LOADED HINT-RATIO GRID. LOWER IS BETTER FOR ALL.

Train seg.	Eval seg.	MSE $\downarrow$	LPIPS $\downarrow$	DreamSim $\downarrow$
DanbooRegion	Felzenszwalb	0.0160 ± 0.0045	0.1960 ± 0.0469	0.0943 ± 0.0221
DanbooRegion	DanbooRegion	0.0172 ± 0.0049	0.2045 ± 0.0470	0.0990 ± 0.0223
DanbooRegion	SLIC	0.0269 ± 0.0044	0.2756 ± 0.0540	0.1554 ± 0.0331
SLIC	Felzenszwalb	0.0203 ± 0.0059	0.2182 ± 0.0445	0.1012 ± 0.0189
SLIC	DanbooRegion	0.0209 ± 0.0061	0.2321 ± 0.0487	0.1104 ± 0.0209
SLIC	SLIC	0.0269 ± 0.0063	0.2370 ± 0.0436	0.1214 ± 0.0214
Felzenszwalb	Felzenszwalb [†]	0.0149 ± 0.0050	0.1829 ± 0.0472	0.0839 ± 0.0212
Felzenszwalb	DanbooRegion	0.0164 ± 0.0051	0.1999 ± 0.0487	0.0945 ± 0.0225
Felzenszwalb	SLIC	0.0243 ± 0.0049	0.2615 ± 0.0563	0.1397 ± 0.0324

[†] The Felz/Felz row is taken from the grid-sensitivity table (Table I, Front-loaded grid), which uses the 96-channel DetFill checkpoint also used for the scribble results in Table II of the main paper; because the other rows come from models trained under each segmenter, its absolute values are not directly comparable to the other rows.

precisely the reproducibility property our protocol is designed to provide. By construction the two methods coincide at the endpoints ($\alpha=0\%$ with no hint and $\alpha=100\%$ with all available stored hints included); the gap is concentrated at the intermediate ratios. Across every metric the seed-to-seed variation of random selection is far smaller than the sort vs random gap, so the HAUC ranking that prefers size-sort is stable under stochastic hint sampling. Figure 7 visualises three representative metrics (MSE, LPIPS, and DreamSim) from the same data for ease of reading; the numerical values are in Table IV. This experiment quantifies seed sensitivity and the selection-rule effect for a single model (DetFill); it does not directly measure cross-model rank reversals across random seeds (see main Tables II–III for the cross-model comparison under the two selection protocols). Unlike this experiment, Table III of the main paper uses one fixed, area-independent region order per image. The standard deviations of that table are across the three sketch sources, not hint-selection seeds.

V. DETAILS OF PROPOSED DETFILL

A. Model Architecture

The architecture is based on a U-Net architecture of guided-diffusion [11] with a 256×256 output. Following guided-diffusion [11], we use up to 96 base channels, the largest setting that fits our memory capacity; we confirm that the colorization quality degrades substantially when the number of base channels is too small (e.g., $C=32$). The scribble model uses 96 base channels, and the dot model uses 64 (a 96-channel dot model did not improve the results in our experiments).

Table V gives an overview of the DetFill U-Net backbone, while Tables VI and VII list the exact layer-by-layer configurations of the ResBlocks and the AttentionBlock, respectively. Our U-Net is built by stacking ResBlocks at each resolution in both the encoder and decoder, with a single self-attention module inserted at the bottleneck. The 5-channel diffusion state stacks the line art (1 channel), the hint-location mask (1 channel), and the RGB color image (3 channels); the 5-channel conditional state replaces the RGB image with the

masked RGB hints. The U-Net input concatenates the current state with the conditional state (10 channels) and predicts the 5-channel objective, and the final colorization is read from the RGB channels of the reconstructed state.

Each ResBlock proceeds as follows:

- 1) Apply GroupNorm $\rightarrow$ SiLU $\rightarrow 3 \times 3$ Conv to the input feature map.
- 2) (Optionally) downsample or upsample both the newly produced features and the original input via identity, average pooling, Upsample operation.
- 3) Inject the time embedding: embed the scalar timestep via SiLU $\rightarrow$ Linear to match the block’s output channels, then broadcast-add it to the feature map.
- 4) Apply GroupNorm $\rightarrow$ SiLU $\rightarrow$ Dropout($p=0$) $\rightarrow 3 \times 3$ Conv.
- 5) Form the residual connection by adding the final conv output to the input via identity or a 1×1 projection.

This design, borrowed from the Guided Diffusion code-base [11], lets us modulate each block by the diffusion timestep without increasing spatial dimensions.

The single AttentionBlock at the bottleneck is implemented as:

- 1) GroupNorm over the bottleneck feature map.
- 2) Conv1d (in = C , out = $3C$) to project into queries, keys, and values.
- 3) QKVAttentionLegacy: multi-head self-attention applied to those projections.
- 4) Conv1d (in = C , out = C) to project the concatenated heads back to C channels.

where C is the bottleneck channel dimension. By combining local processing in the ResBlocks with a global attention pass at the centre, we achieve both high-fidelity detail synthesis and long-range consistency.

VI. ADDITIONAL DETAILS OF FORMATIVE STUDY

Ethics statement. Both the formative study (main Sec. III) and the perceptual user study (main Sec. VII) are non-invasive, non-interventional studies with healthy adult volunteers; under the university’s research-ethics guidelines, studies of this type

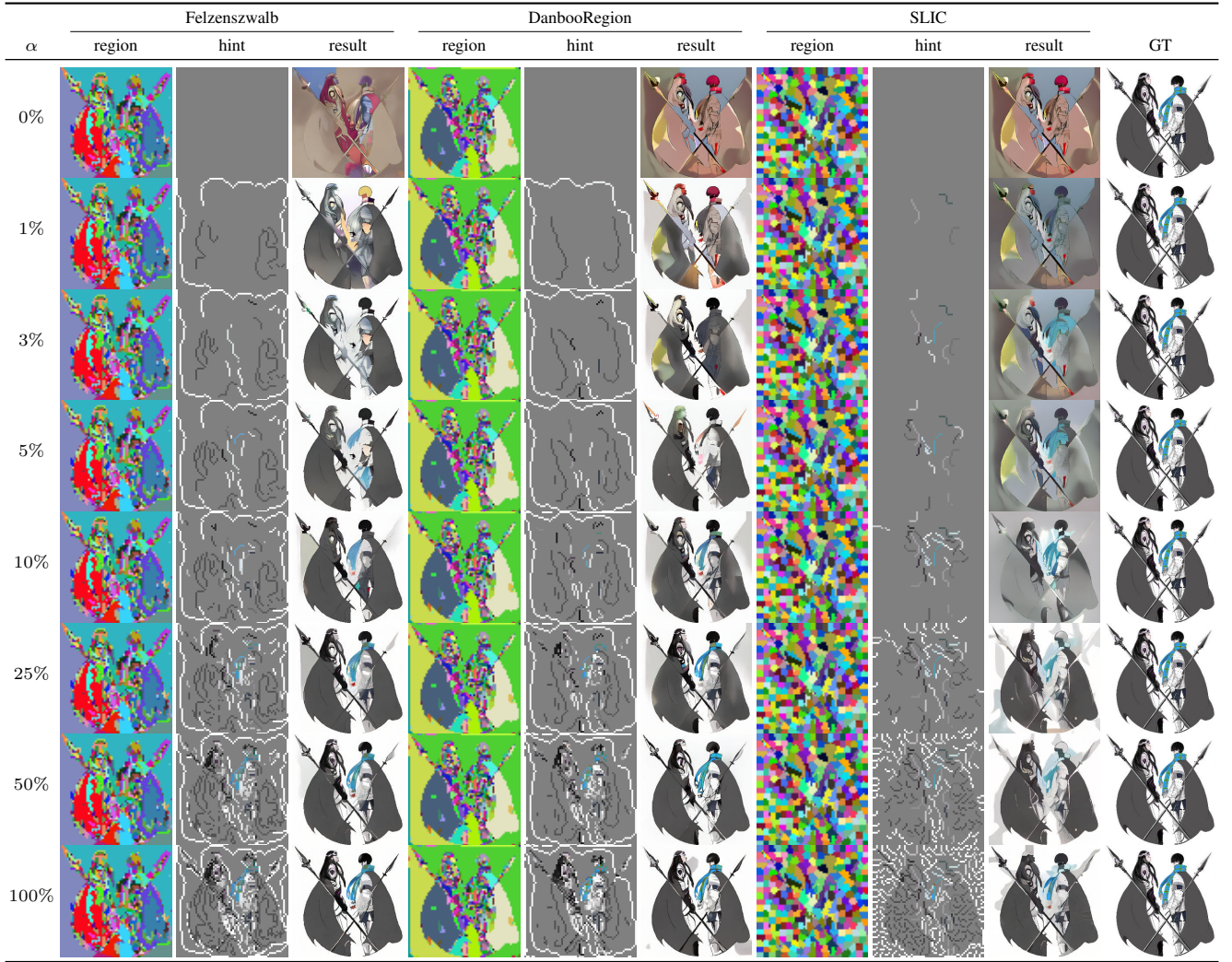


Fig. 5. **Robustness to the region segmenter — all hint ratios (example 1019016).** A single model (DetFill trained on the DanbooRegion segmenter) evaluated under three segmentation backbones, with sketch simplification (Simo-Serra et al., “Mastering Sketching”) as the line-art input. The hint ratio α increases downward; for each evaluation segmenter (Felzenszwalb, DanbooRegion, SLIC) we show the region map, the per- α color hint, and the resulting colorization, with the ground truth (GT) repeated at every ratio. Only the hint-generation segmenter varies across columns; the colorization model is fixed. As α grows the hints densify and the three segmenters converge to similar colorizations.

do not require formal ethics-committee review. All participants provided informed consent (via a signed consent form in the perceptual study) and were fairly compensated for their time.

A. Strategies of colorization from survey

From post-survey results from 16 participants, we summarize the strategies of colorization as follows:

- **Color Harmony and Balance:** Many participants aimed to maintain an overall balance in color, ensuring that the image looked cohesive and visually pleasing.
- **Reference vs. Creative Choice:** When a reference image or correct colors were available, participants tried to replicate them accurately. Without a reference, they used their creativity to apply colors freely or in a way that created a certain mood or effect.
- **Strategic Workflow:** Participants often began coloring from large or visually dominant areas, layering colors gradually and refining details later. Some also considered

the overall composition by occasionally stepping back to view the image holistically.

- **Efficiency and Practicality:** Especially among hobbyists, there was an emphasis on finishing the task efficiently while maintaining acceptable quality, using tools like bucket fill or brush depending on the situation.

As mentioned in the main paper, users often employ a strategy of colorizing from large regions to smaller ones. In addition, many participants pay attention to overall color balance to make the colorized images visually appealing. Although our proposed metric does not explicitly account for color adjustment or aesthetic design, such aspects can be partially captured by existing perceptual metrics such as LPIPS.

B. User-wise Final Colorization Results

Fig. 8, 9 visualize the final colorization results at the formative study. Each user exhibits unique characteristics in their approach to colorization. For example, some users focus on coloring the background in non-reference cases, while

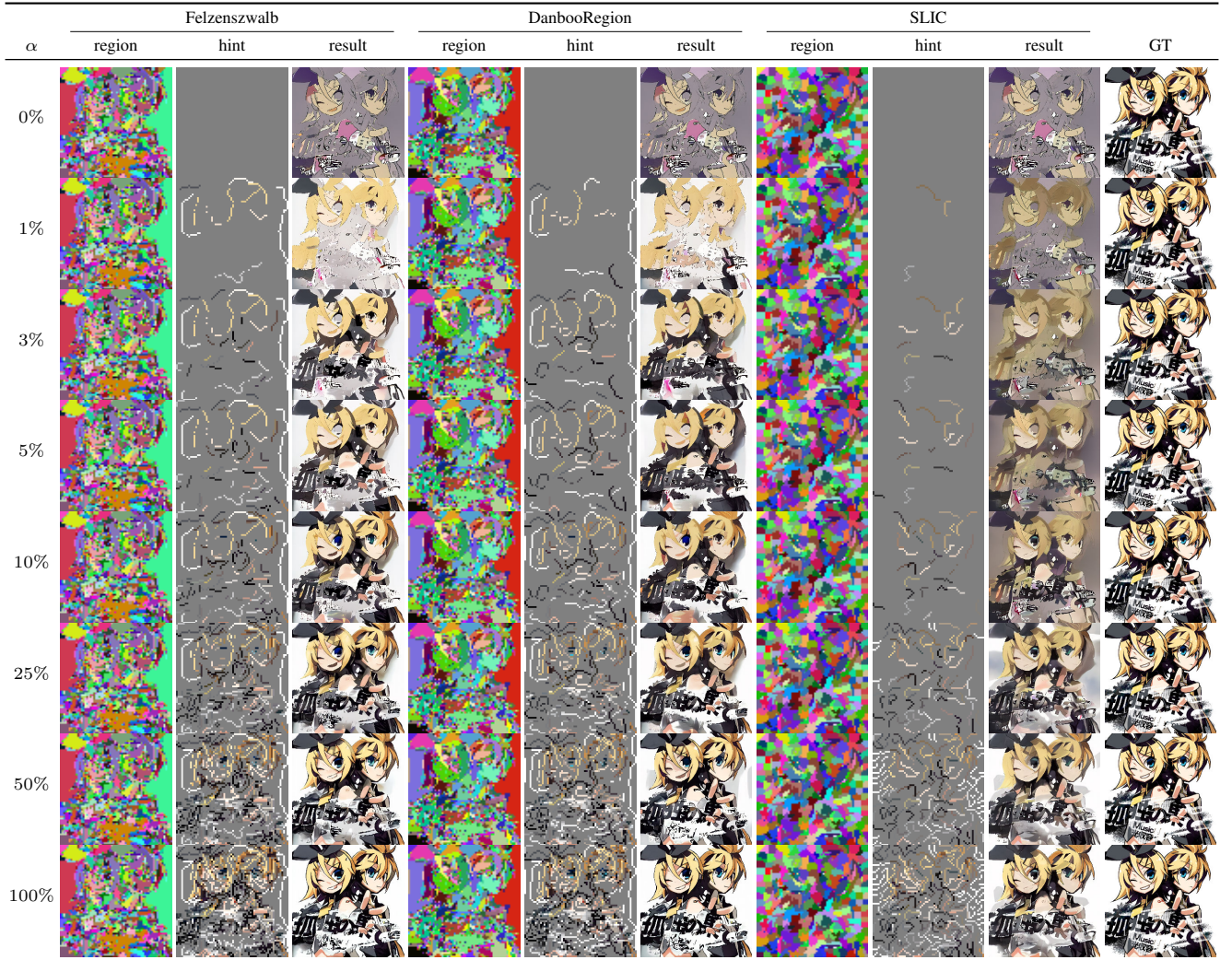


Fig. 6. **Robustness to the region segmenter — all hint ratios (example 1023016).** A single model (DetFill trained on the DanbooRegion segmenter) evaluated under three segmentation backbones, with sketch simplification (Simo-Serra et al., “Mastering Sketching”) as the line-art input. The hint ratio α increases downward; for each evaluation segmenter (Felzenszwalb, DanbooRegion, SLIC) we show the region map, the per- α color hint, and the resulting colorization, with the ground truth (GT) repeated at every ratio. Only the hint-generation segmenter varies across columns; the colorization model is fixed. As α grows the hints densify and the three segmenters converge to similar colorizations.

others concentrate on specific areas or make tone adjustments. Overall, we believe there is no significant difference in the final colorization results such as the quality of the colorization—between the reference and non-reference conditions.

VII. EVALUATION WITH NATURAL IMAGE

Tables VIII and IX show the results in which we replace the Danbooru2021 [4] dataset with ImageNet [12].

From these results, we confirm a similar overall tendency: DetFill remains the best on all pixel-level metrics (MSE, PSNR, SSIM, and LPIPS), whereas the ColorizeDiffusion variants fine-tuned with the original 7-epoch schedule (7ep) surpass it on several semantic-feature metrics, especially for dot hints (ColDiffv1 on OpenCLIP and DINO under deterministic hints, with additional semantic-metric cases under random sampling), while DetFill remains best on every metric for deterministic scribble hints.

VIII. IMAGE VISUALIZATION

We also visualize the deterministic hint sampling for illustrations in Figures 10, 11 and 12, 13, as well as for natural images in Figures 14, 15 and 16, 17. Additionally, we show the random hint sampling for illustrations in Figures 18, 19 and 20, 21, and for natural images in Figures 22, 23 and 24, 25. The proposed DetFill method consistently produces superior colorization results across different hint sampling ratios. On ImageNet, the ColorizeDiffusion variants fine-tuned with the original 7-epoch schedule (7ep) surpass DetFill on several semantic-feature metrics, especially for dot hints (ColDiffv1 on OpenCLIP and DINO under deterministic hints, with additional semantic-metric cases under random sampling; Tables VIII and IX), consistent with their large-scale natural-image pretraining, whereas DetFill remains the best on all pixel-level metrics (MSE, PSNR, SSIM, and LPIPS), on every metric for deterministic scribble hints, and on every metric against the ColorizeDiffusion variants for illustrations, the primary target domain of our protocol.

TABLE IV

PER-RATIO METRICS AND HINT-AUC COMPARING THE PROPOSED SIZE-SORT HINT SELECTION AGAINST UNIFORMLY RANDOM REGION SELECTION (96-CHANNEL DetFill trained on FELZENSZWALB, SKETCHKERAS LINE ART, $N=300$ TEST IMAGES). EACH METRIC IS REPORTED AT EVERY HINT RATIO $\alpha \in \{0, 1, 3, 5, 10, 25, 50, 100\}\%$, TOGETHER WITH THE HAUC TRAPEZOIDAL INTEGRAL OVER THE SAME GRID NORMALISED BY 100. FOR THE RANDOM ROW EACH CELL IS THE MEAN $\pm$ SAMPLE SD OVER 10 INDEPENDENT HINT-SAMPLING SEEDS (THE SEED-SENSITIVITY CHECK); THE Δ (SORT-RAND) ROW REPORTS THE PER-COLUMN DIFFERENCE. THE SIZE-SORT ROW REPORTS A SINGLE VALUE PER CELL WITHOUT A STANDARD DEVIATION BECAUSE THE PROPOSED SIZE-SORT SELECTION IS DETERMINISTIC AND HAS NO SEED-TO-SEED VARIATION BY CONSTRUCTION. AT $\alpha=0\%$ NO HINT IS PROVIDED AND AT $\alpha=100\%$ ALL AVAILABLE STORED HINTS ARE INCLUDED, SO THE TWO METHODS COINCIDE AT THE GRID ENDPOINTS BY CONSTRUCTION ($\Delta=0$). ARROWS MARK THE BETTER DIRECTION; BOLD MARKS THE BETTER OF THE TWO METHODS PER COLUMN. THE PROPOSED SIZE-SORT IMPROVES EVERY METRIC IN HAUC, AND THE SEED-TO-SEED VARIATION OF RANDOM SELECTION IS FAR SMALLER THAN THE SORT VS RANDOM GAP.

Metric	Method	hint ratio α (%)								HAUC
		0	1	3	5	10	25	50	100	
MSE $\downarrow$	size-sort	0.1003	0.0501	0.0310	0.0239	0.0166	0.0113	0.0089	0.0075	0.0119
	random (10 seeds)	0.1003	0.0706 $\pm$ 0.0017	0.0535 $\pm$ 0.0011	0.0439 $\pm$ 0.0006	0.0316 $\pm$ 0.0005	0.0183 $\pm$ 0.0003	0.0117 $\pm$ 0.0002	0.0075	0.0172 $\pm$ 0.0001
	Δ (sort-rand)	0	-0.0205	-0.0225	-0.0201	-0.0150	-0.0070	-0.0027	0	-0.0054
LPIPS $\downarrow$	size-sort	0.4405	0.3315	0.2673	0.2357	0.1954	0.1496	0.1206	0.0991	0.1402
	random (10 seeds)	0.4405	0.3592 $\pm$ 0.0028	0.3062 $\pm$ 0.0021	0.2764 $\pm$ 0.0012	0.2322 $\pm$ 0.0010	0.1721 $\pm$ 0.0009	0.1316 $\pm$ 0.0005	0.0991	0.1551 $\pm$ 0.0004
	Δ (sort-rand)	0	-0.0277	-0.0389	-0.0407	-0.0368	-0.0225	-0.0109	0	-0.0149
DreamSim $\downarrow$	size-sort	0.3463	0.2518	0.1709	0.1346	0.0969	0.0652	0.0499	0.0399	0.0651
	random (10 seeds)	0.3463	0.2575 $\pm$ 0.0022	0.1934 $\pm$ 0.0018	0.1632 $\pm$ 0.0012	0.1222 $\pm$ 0.0010	0.0780 $\pm$ 0.0010	0.0546 $\pm$ 0.0002	0.0399	0.0734 $\pm$ 0.0003
	Δ (sort-rand)	0	-0.0056	-0.0225	-0.0286	-0.0253	-0.0128	-0.0047	0	-0.0084
PSNR $\uparrow$ (dB)	size-sort	10.43	13.74	15.81	16.96	18.57	20.34	21.43	22.30	20.71
	random (10 seeds)	10.43	12.27 $\pm$ 0.08	13.56 $\pm$ 0.07	14.41 $\pm$ 0.05	15.85 $\pm$ 0.04	18.29 $\pm$ 0.05	20.32 $\pm$ 0.03	22.30	19.45 $\pm$ 0.02
	Δ (sort-rand)	0	+1.47	+2.26	+2.55	+2.72	+2.05	+1.11	0	+1.26
SSIM $\uparrow$	size-sort	0.5317	0.6213	0.6691	0.6940	0.7278	0.7705	0.8015	0.8284	0.7842
	random (10 seeds)	0.5317	0.5812 $\pm$ 0.0025	0.6206 $\pm$ 0.0020	0.6456 $\pm$ 0.0013	0.6864 $\pm$ 0.0008	0.7477 $\pm$ 0.0010	0.7921 $\pm$ 0.0006	0.8284	0.7687 $\pm$ 0.0004
	Δ (sort-rand)	0	+0.0401	+0.0485	+0.0483	+0.0414	+0.0227	+0.0095	0	+0.0155
OpenCLIP $\uparrow$	size-sort	0.8767	0.9050	0.9286	0.9397	0.9518	0.9629	0.9686	0.9729	0.9636
	random (10 seeds)	0.8767	0.9015 $\pm$ 0.0013	0.9209 $\pm$ 0.0008	0.9303 $\pm$ 0.0008	0.9434 $\pm$ 0.0007	0.9584 $\pm$ 0.0006	0.9670 $\pm$ 0.0003	0.9729	0.9608 $\pm$ 0.0003
	Δ (sort-rand)	0	+0.0035	+0.0077	+0.0094	+0.0084	+0.0044	+0.0015	0	+0.0028
DINO $\uparrow$	size-sort	0.8992	0.9229	0.9367	0.9434	0.9521	0.9612	0.9664	0.9709	0.9627
	random (10 seeds)	0.8992	0.9122 $\pm$ 0.0012	0.9231 $\pm$ 0.0008	0.9304 $\pm$ 0.0007	0.9411 $\pm$ 0.0005	0.9554 $\pm$ 0.0007	0.9646 $\pm$ 0.0003	0.9709	0.9588 $\pm$ 0.0003
	Δ (sort-rand)	0	+0.0107	+0.0136	+0.0130	+0.0111	+0.0058	+0.0018	0	+0.0038

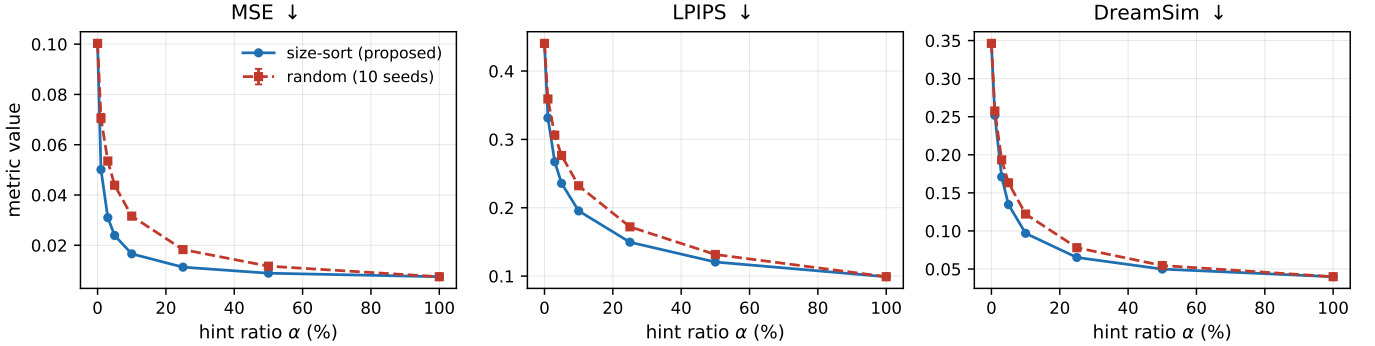


Fig. 7. Per-ratio MSE, LPIPS, and DreamSim for the proposed size-sort hint selection versus uniformly random region selection (mean $\pm$ sample SD over 10 hint-sampling seeds; 96-channel DetFill trained on Felzenszwalb, SketchKeras line art, $N=300$ test images). The curves visualize the same data as Table IV: the two methods coincide at the endpoints by construction, the gap concentrates at the intermediate ratios, and the seed-to-seed variation of random selection (error bars) is far smaller than the sort-vs-random gap. The size-sort curve carries no error bars because the proposed selection is deterministic.

IX. SKETCHWISE COLORIZATION

Figure 26 visualize the DetFill’s colorization results by the same hints over the different sketch inputs. We can verify the colorization difference from eyes and hair styles. The results indicate that using different sketches is important to comprehensively evaluate the colorization model.

X. ADDITIONAL TRAINING COMPARISONS

Beyond the common 200-epoch training protocol used for all baselines in the main paper, we report two additional training comparisons: (i) ColorizeDiffusion fine-tuned with the authors’ original schedule, and (ii) Diffusart retrained with

our deterministic hints, isolating the training protocol from the model architecture.

Regarding the choice of modern baselines: among recent reference-conditioned systems, the ColorizeDiffusion v2 paper already reports that ControlNet-based pipelines with IP-Adapter or T2I-Adapter conditioning perform substantially worse than ColorizeDiffusion v2; as of July 2026, neither Cobra nor MagicColor provides a complete, documented training recipe, which retraining under our common protocol requires; and LVCD is a video-colorization model outside the single-image scope (main Sec. VIII). We therefore adopt ColorizeDiffusion v1/v2 as the modern reference-conditioned



Fig. 8. **User colorization results:** Each user draws colors in the non-reference case on the left side and the reference case on the right side.

baselines.

Original fine-tuning schedule (7ep). Both ColorizeDiffusion variants start from the officially released checkpoints. In addition to the 200-epoch protocol reported in the main paper (200ep), we fine-tune both models with the authors’ original schedule (7 epochs, a flat learning rate of 1×10^{-5} , and an effective batch size of 128) at 256×256 , denoted 7ep, and evaluate them under the identical protocol. Comparing the two schedules, the original 7-epoch schedule generally improves the perceptual scores of ColorizeDiffusion over the 200-epoch schedule under deterministic hints (most consistently DreamSim; Table X versus Table II of the main paper) and also under random sampling (Table XI versus Table III of the main paper; e.g., DreamSim $0.652 \rightarrow 0.219$ for ColDiffv1 with dot hints), with small exceptions for ColDiffv2 (notably LPIPS with both hint types and OpenCLIP/DINO with scribble hints). In both schedules, ColorizeDiffusion stays below DetFill on every metric on the illustration benchmark, indicating that the observed margin is not an artifact of under-trained baselines. On natural images, the fine-tuned baselines surpass DetFill

on several semantic-feature metrics for dot hints (Tables VIII and IX), consistent with their large-scale natural-image pre-training.

Architecture ablation (Diffusart-retrain). To isolate the contribution of the deterministic-hint training protocol from the model architecture, we retrain Diffusart under exactly the same 200-epoch protocol and deterministic region-based hints as DetFill, replacing its original random hint sampling (denoted Diffusart-retrain). As shown in Table X, this training protocol lifts Diffusart close to DetFill: DetFill is better on most metrics for both hint types (e.g., scribble PSNR 19.723 vs. 19.457 and MSE 0.015 vs. 0.016), while Diffusart-retrain is slightly better on scribble LPIPS (0.177 vs. 0.183) and dot DreamSim (0.124 vs. 0.134), and the two are tied on scribble DreamSim (0.084). These results indicate that the improvements primarily stem from the deterministic region-based hint generation and training protocol rather than from a specific architecture, and that the protocol transfers across architectures.

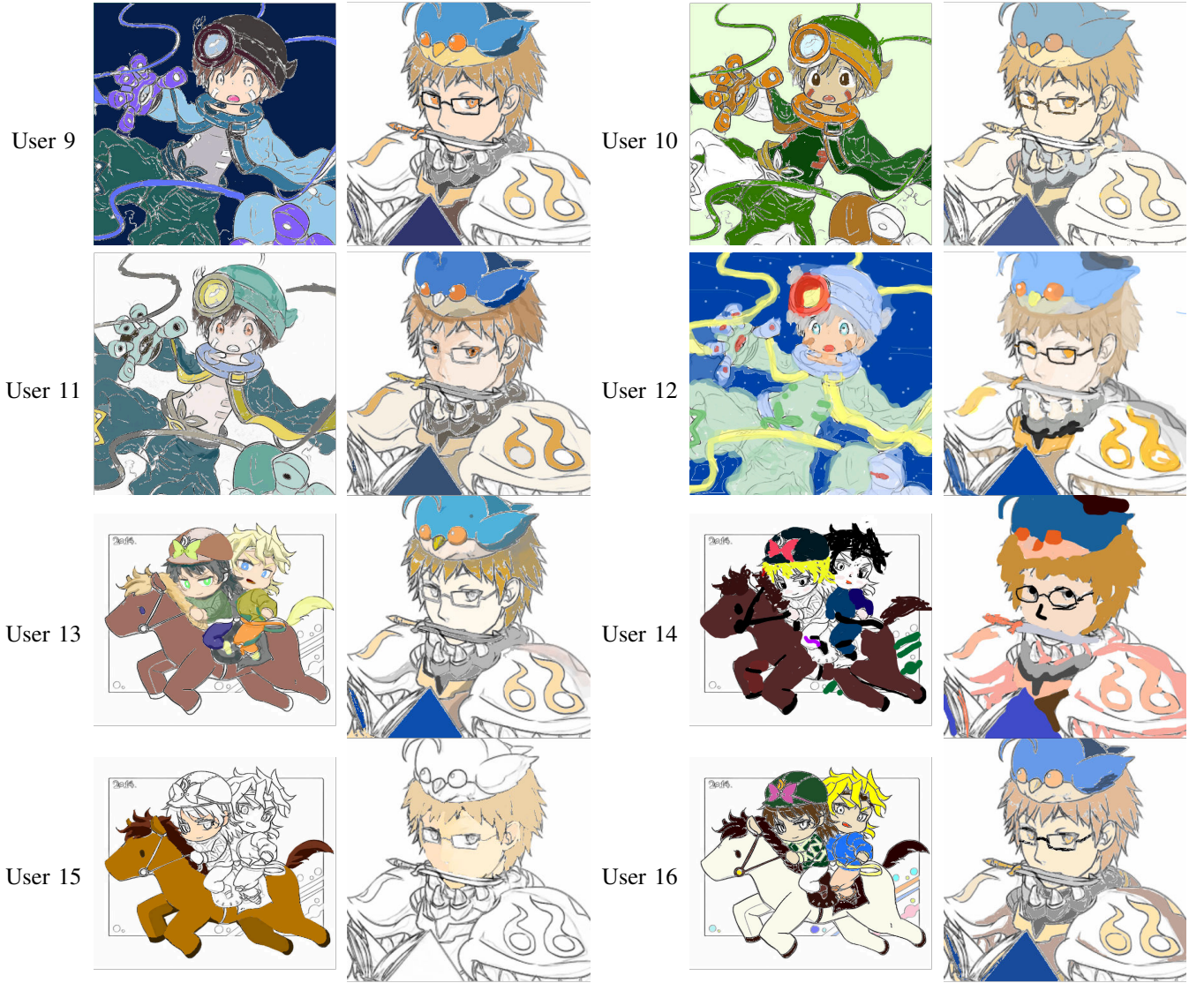


Fig. 9. **User colorization results:** Each user draws colors in the non-reference case on the left side and the reference case on the right side.

TABLE V

U-NET ARCHITECTURE FOR DETFILL WITH 256×256 INPUTS. SELF-ATTENTION (“ATTENTIONBLOCK”) APPEARS ONLY ONCE IN THE BOTTLENECK (64×64). THE 96-CHANNEL (SCRIBBLE) CONFIGURATION IS SHOWN; THE DOT MODEL USES 64 BASE CHANNELS (SEE THE TEXT).

Stage	Layer	Channels	Feature Map Size
Input	-	10	256×256
	Conv 3×3	96	256×256
Downsample	ResBlock $\times 2$	96	256×256
	ResBlock $\times 2$	384	128×128
	ResBlock $\times 2$	768	64×64
Bottleneck	ResBlock	768	64×64
	AttentionBlock	768	64×64
	ResBlock	768	64×64
Upsample	ResBlock $\times 2$	768	64×64
	ResBlock $\times 2$	384	128×128
	ResBlock $\times 2$	96	256×256
Output	Conv 3×3	5	256×256

TABLE VI
DETAILED RESBLOCK STRUCTURE IN DETFILL'S U-NET

Step	Layer	Input	Output	Channels
Input layers	GroupNorm32	x	$u = \text{GN}(x)$	input channels
	SiLU	u	$v = \text{SiLU}(u)$	input channels
	Conv 3×3 (pad=1)	v	$h_1 = \text{Conv}(v)$	output channels
h processing	Identity / Downsample / Upsample	h_1	h_2	output channels
x processing	Identity / Downsample / Upsample	x	x_2	input channels
Embedding Layers	SiLU on time-embed e	e	$e_1 = \text{SiLU}(e)$	—
	Linear(in=embedding channels, out=output channels)	e_1	e'	output channels
Output layers	GroupNorm32	$h_2 + e'$	$w = \text{GN}(h_2 + e')$	—
	SiLU	w	$z = \text{SiLU}(w)$	—
	Dropout(p=0)	z	z'	—
	Conv 3×3 (pad=1)	z'	$h_4 = \text{Conv}(z')$	output channels
Skip Connection	Identity (or 1×1 Conv if ch. differ)	x_2	$\tilde{x}$	output channels
Add	Add residual	$\tilde{x}, h_4$	$y = \tilde{x} + h_4$	output channels

TABLE VII
DETAILED ATTENTIONBLOCK STRUCTURE IN DETFILL'S U-NET

Operation	Description	Channels / Params
GroupNorm32	normalization	input channels
Conv1d	Q/K/V projection	$3 \times$ input channels
QKVAttentionLegacy	attention mechanism	—
Conv1d	output projection	input channels

TABLE VIII

HINT-AUC OVER THE DETERMINISTIC HINTS BY REGION-BASED HINT SAMPLING WITH NATURAL IMAGE DATASET (IMAGENET): THE RESULT INDICATES THAT DIFFERENT EVALUATION SETTINGS OF HINT-AUC PRODUCE DIFFERENT SCORES AND TENDENCIES OVER THE MODES. EACH SCORE IS THE MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THE THREE SKETCH SOURCES (XDOG, SKETCH SIMPLIFICATION, AND SKETCHKERAS; $n = 3$, DIVISOR $n - 1$). 200EP/7EP: COLORIZEDIFFUSION FINE-TUNED FROM THE OFFICIAL WEIGHTS WITH OUR 200-EPOCH PROTOCOL OR THE AUTHORS' ORIGINAL 7-EPOCH SCHEDULE.

Hint	Model	MSE ($\downarrow$)	PSNR ($\uparrow$)	SSIM ($\uparrow$)	Hint-AUC LPIPS ($\downarrow$)	OpenCLIP ($\uparrow$)	DINO ($\uparrow$)	DreamSim ($\downarrow$)
Dot	PaintsTorch [13]	0.047 $\pm$ 0.005	12.968 $\pm$ 0.25	0.421 $\pm$ 0.052	0.520 $\pm$ 0.045	0.753 $\pm$ 0.021	0.718 $\pm$ 0.044	0.590 $\pm$ 0.049
	Diffusart [14]	0.052 $\pm$ 0.007	12.532 $\pm$ 0.31	0.453 $\pm$ 0.062	0.464 $\pm$ 0.046	0.775 $\pm$ 0.030	0.721 $\pm$ 0.049	0.516 $\pm$ 0.057
	ColDiffv1 (200ep) [15]	0.062 $\pm$ 0.012	12.971 $\pm$ 0.99	0.330 $\pm$ 0.094	0.486 $\pm$ 0.084	0.767 $\pm$ 0.052	0.719 $\pm$ 0.083	0.525 $\pm$ 0.124
	ColDiffv2 (200ep) [16]	0.046 $\pm$ 0.009	14.174 $\pm$ 0.84	0.379 $\pm$ 0.073	0.457 $\pm$ 0.062	0.783 $\pm$ 0.042	0.729 $\pm$ 0.069	0.466 $\pm$ 0.089
	ColDiffv1 (7ep) [15]	0.071 $\pm$ 0.012	12.332 $\pm$ 0.86	0.383 $\pm$ 0.051	0.423 $\pm$ 0.056	0.846$\pm$0.024	0.818$\pm$0.036	0.345 $\pm$ 0.050
	ColDiffv2 (7ep) [16]	0.067 $\pm$ 0.009	12.533 $\pm$ 0.66	0.366 $\pm$ 0.045	0.441 $\pm$ 0.052	0.830 $\pm$ 0.024	0.791 $\pm$ 0.035	0.361 $\pm$ 0.046
	DetFill	0.027$\pm$0.006	15.335$\pm$0.39	0.542$\pm$0.078	0.326$\pm$0.062	0.845 $\pm$ 0.038	0.800 $\pm$ 0.061	0.334$\pm$0.075
Scribble	PaintsTorch [13]	0.110 $\pm$ 0.023	9.384 $\pm$ 0.35	0.403 $\pm$ 0.052	0.490 $\pm$ 0.033	0.756 $\pm$ 0.029	0.727 $\pm$ 0.056	0.553 $\pm$ 0.067
	Diffusart [14]	0.035 $\pm$ 0.008	14.014 $\pm$ 0.47	0.518 $\pm$ 0.071	0.347 $\pm$ 0.054	0.828 $\pm$ 0.036	0.778 $\pm$ 0.059	0.384 $\pm$ 0.069
	ColDiffv1 (200ep) [15]	0.053 $\pm$ 0.009	13.477 $\pm$ 0.84	0.390 $\pm$ 0.082	0.473 $\pm$ 0.094	0.776 $\pm$ 0.055	0.735 $\pm$ 0.085	0.503 $\pm$ 0.127
	ColDiffv2 (200ep) [16]	0.044 $\pm$ 0.008	14.544 $\pm$ 0.94	0.390 $\pm$ 0.078	0.426 $\pm$ 0.063	0.801 $\pm$ 0.041	0.753 $\pm$ 0.067	0.423 $\pm$ 0.085
	ColDiffv1 (7ep) [15]	0.065 $\pm$ 0.010	12.764 $\pm$ 0.77	0.394 $\pm$ 0.047	0.419 $\pm$ 0.054	0.852 $\pm$ 0.021	0.827 $\pm$ 0.031	0.327 $\pm$ 0.044
	ColDiffv2 (7ep) [16]	0.054 $\pm$ 0.008	13.487 $\pm$ 0.72	0.406 $\pm$ 0.041	0.414 $\pm$ 0.047	0.847 $\pm$ 0.020	0.814 $\pm$ 0.029	0.326 $\pm$ 0.040
	DetFill	0.017$\pm$0.004	17.531$\pm$0.48	0.596$\pm$0.074	0.263$\pm$0.053	0.872$\pm$0.030	0.836$\pm$0.052	0.270$\pm$0.060

TABLE IX

HINT-AUC OVER THE DETERMINISTIC HINTS BY RANDOM SAMPLING WITH NATURAL IMAGE DATASET (IMAGENET): THE RESULT INDICATES THAT DIFFERENT EVALUATION SETTINGS OF HINT-AUC PRODUCE DIFFERENT SCORES AND TENDENCIES OVER THE MODES. EACH SCORE IS THE MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THE THREE SKETCH SOURCES (XDOG, SKETCH SIMPLIFICATION, AND SKETCHKERAS; $n = 3$, DIVISOR $n - 1$). 200EP/7EP: COLORIZEDIFFUSION FINE-TUNED FROM THE OFFICIAL WEIGHTS WITH OUR 200-EPOCH PROTOCOL OR THE AUTHORS' ORIGINAL 7-EPOCH SCHEDULE.

Hint	Model	MSE ($\downarrow$)	PSNR ($\uparrow$)	SSIM ($\uparrow$)	Hint-AUC LPIPS ($\downarrow$)	OpenCLIP ($\uparrow$)	DINO ($\uparrow$)	DreamSim ($\downarrow$)
Dot	PaintsTorch [13]	0.055 $\pm$ 0.007	12.254 $\pm$ 0.27	0.408 $\pm$ 0.053	0.530 $\pm$ 0.045	0.749 $\pm$ 0.021	0.712 $\pm$ 0.044	0.600 $\pm$ 0.048
	Diffusart [14]	0.063 $\pm$ 0.009	11.730 $\pm$ 0.29	0.427 $\pm$ 0.061	0.491 $\pm$ 0.045	0.765 $\pm$ 0.029	0.707 $\pm$ 0.048	0.541 $\pm$ 0.056
	ColDiffv1 (200ep) [15]	0.059 $\pm$ 0.008	13.112 $\pm$ 0.71	0.358 $\pm$ 0.076	0.480 $\pm$ 0.076	0.769 $\pm$ 0.052	0.721 $\pm$ 0.081	0.521 $\pm$ 0.125
	ColDiffv2 (200ep) [16]	0.053 $\pm$ 0.009	13.727 $\pm$ 0.78	0.374 $\pm$ 0.071	0.468 $\pm$ 0.061	0.780 $\pm$ 0.041	0.726 $\pm$ 0.068	0.472 $\pm$ 0.088
	ColDiffv1 (7ep) [15]	0.073 $\pm$ 0.012	12.272 $\pm$ 0.84	0.384 $\pm$ 0.050	0.427 $\pm$ 0.055	0.847$\pm$0.024	0.819$\pm$0.035	0.343$\pm$0.049
	ColDiffv2 (7ep) [16]	0.072 $\pm$ 0.009	12.283 $\pm$ 0.66	0.365 $\pm$ 0.045	0.445 $\pm$ 0.052	0.829 $\pm$ 0.024	0.790 $\pm$ 0.035	0.362 $\pm$ 0.047
	DetFill	0.038$\pm$0.007	13.996$\pm$0.37	0.516$\pm$0.078	0.362$\pm$0.064	0.830 $\pm$ 0.039	0.782 $\pm$ 0.063	0.373 $\pm$ 0.077
Scribble	PaintsTorch [13]	0.116 $\pm$ 0.029	9.081 $\pm$ 0.43	0.400 $\pm$ 0.055	0.495 $\pm$ 0.037	0.754 $\pm$ 0.029	0.724 $\pm$ 0.056	0.562 $\pm$ 0.068
	Diffusart [14]	0.044 $\pm$ 0.011	13.119 $\pm$ 0.50	0.497 $\pm$ 0.075	0.370 $\pm$ 0.056	0.821 $\pm$ 0.037	0.767 $\pm$ 0.060	0.404 $\pm$ 0.071
	ColDiffv1 (200ep) [15]	0.057 $\pm$ 0.008	13.261 $\pm$ 0.78	0.372 $\pm$ 0.080	0.482 $\pm$ 0.090	0.773 $\pm$ 0.054	0.731 $\pm$ 0.085	0.511 $\pm$ 0.126
	ColDiffv2 (200ep) [16]	0.061 $\pm$ 0.011	13.431 $\pm$ 0.95	0.335 $\pm$ 0.085	0.459 $\pm$ 0.066	0.793 $\pm$ 0.042	0.742 $\pm$ 0.067	0.442 $\pm$ 0.086
	ColDiffv1 (7ep) [15]	0.071 $\pm$ 0.012	12.412 $\pm$ 0.84	0.380 $\pm$ 0.050	0.429 $\pm$ 0.056	0.853 $\pm$ 0.022	0.827$\pm$0.032	0.329 $\pm$ 0.045
	ColDiffv2 (7ep) [16]	0.073 $\pm$ 0.012	12.410 $\pm$ 0.82	0.363 $\pm$ 0.049	0.433 $\pm$ 0.051	0.842 $\pm$ 0.021	0.808 $\pm$ 0.031	0.336 $\pm$ 0.043
	DetFill	0.027$\pm$0.006	15.677$\pm$0.48	0.561$\pm$0.079	0.300$\pm$0.059	0.858$\pm$0.033	0.816 $\pm$ 0.058	0.307$\pm$0.068

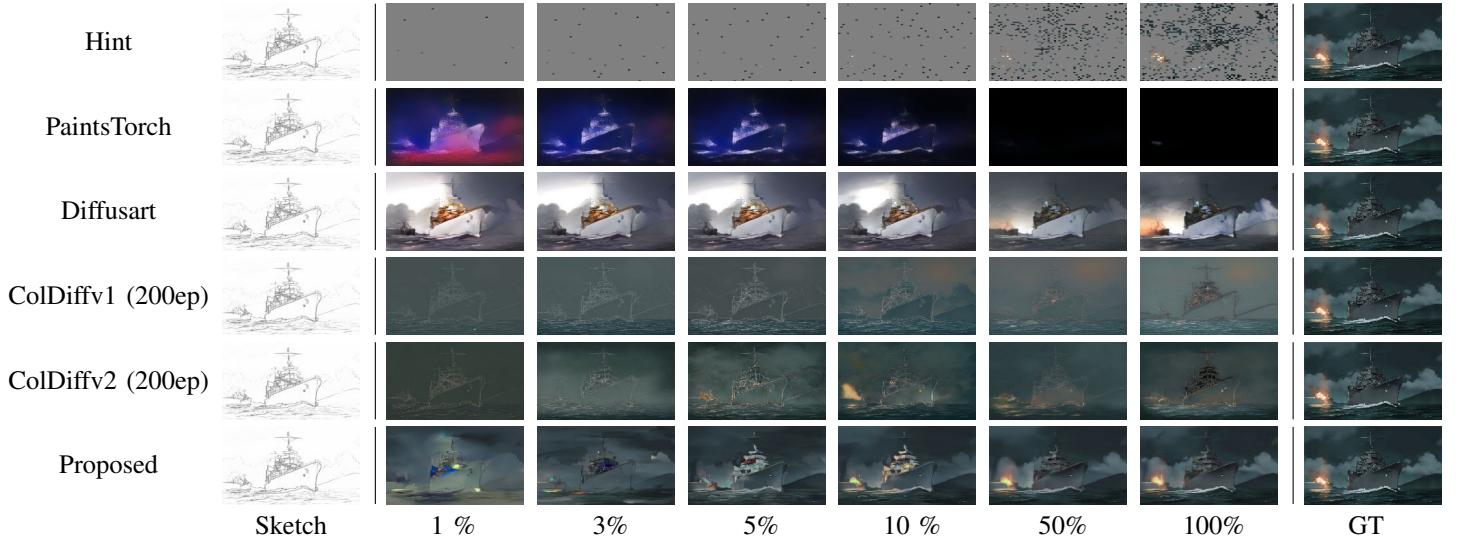


Fig. 10. **Colorization with different hint ratios with deterministic hints by region-based hint sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

TABLE X

ADDITIONAL TRAINING COMPARISONS WITH DETERMINISTIC HINTS BY REGION-BASED HINT SAMPLING ON THE DANBOORU2021 DATASET (HINT-AUC). 7EP: COLORIZEDIFFUSION FINE-TUNED FROM THE OFFICIAL WEIGHTS WITH THE AUTHORS' ORIGINAL 7-EPOCH SCHEDULE (THE CORRESPONDING 200EP RESULTS ARE REPORTED IN TABLE II OF THE MAIN PAPER). DIFFUSART-RETRAIN: DIFFUSART RETRAINED WITH OUR DETERMINISTIC HINTS UNDER THE SAME 200-EPOCH PROTOCOL AS DETFILL. THE DETFILL ROWS ARE REPEATED FROM TABLE II OF THE MAIN PAPER FOR REFERENCE. EACH SCORE IS THE MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THE THREE SKETCH SOURCES (XDOG, SKETCH SIMPLIFICATION, AND SKETCHKERAS; $n = 3$, DIVISOR $n - 1$).

Hint	Model	MSE ($\downarrow$)	PSNR ($\uparrow$)	SSIM ($\uparrow$)	Hint-AUC LPIPS ($\downarrow$)	OpenCLIP ($\uparrow$)	DINO ($\uparrow$)	DreamSim ($\downarrow$)
Dot	ColDiffv1 (7ep) [15]	0.074 $\pm$ 0.010	12.370 $\pm$ 0.62	0.491 $\pm$ 0.041	0.367 $\pm$ 0.034	0.915 $\pm$ 0.008	0.929 $\pm$ 0.010	0.223 $\pm$ 0.021
	ColDiffv2 (7ep) [16]	0.079 $\pm$ 0.009	12.119 $\pm$ 0.56	0.475 $\pm$ 0.034	0.377 $\pm$ 0.027	0.908 $\pm$ 0.008	0.921 $\pm$ 0.008	0.227 $\pm$ 0.017
	Diffusart-retrain	0.027 $\pm$ 0.012	17.156 $\pm$ 2.07	0.658 $\pm$ 0.059	0.232 $\pm$ 0.034	0.936 $\pm$ 0.009	0.925 $\pm$ 0.005	0.124$\pm$0.028
	DetFill (Ours)	0.021$\pm$0.004	17.321$\pm$0.91	0.675$\pm$0.056	0.209$\pm$0.029	0.938$\pm$0.008	0.935$\pm$0.013	0.134 $\pm$ 0.018
Scribble	ColDiffv1 (7ep) [15]	0.061 $\pm$ 0.008	13.273 $\pm$ 0.60	0.522 $\pm$ 0.034	0.334 $\pm$ 0.032	0.923 $\pm$ 0.007	0.935 $\pm$ 0.008	0.193 $\pm$ 0.017
	ColDiffv2 (7ep) [16]	0.060 $\pm$ 0.007	13.388 $\pm$ 0.56	0.502 $\pm$ 0.033	0.342 $\pm$ 0.027	0.915 $\pm$ 0.007	0.925 $\pm$ 0.008	0.202 $\pm$ 0.016
	Diffusart-retrain	0.016 $\pm$ 0.004	19.457 $\pm$ 1.06	0.717 $\pm$ 0.066	0.177$\pm$0.031	0.952 $\pm$ 0.008	0.943 $\pm$ 0.009	0.084$\pm$0.015
	DetFill (Ours)	0.015$\pm$0.005	19.723$\pm$1.41	0.724$\pm$0.086	0.183 $\pm$ 0.047	0.954$\pm$0.010	0.951$\pm$0.011	0.084$\pm$0.021

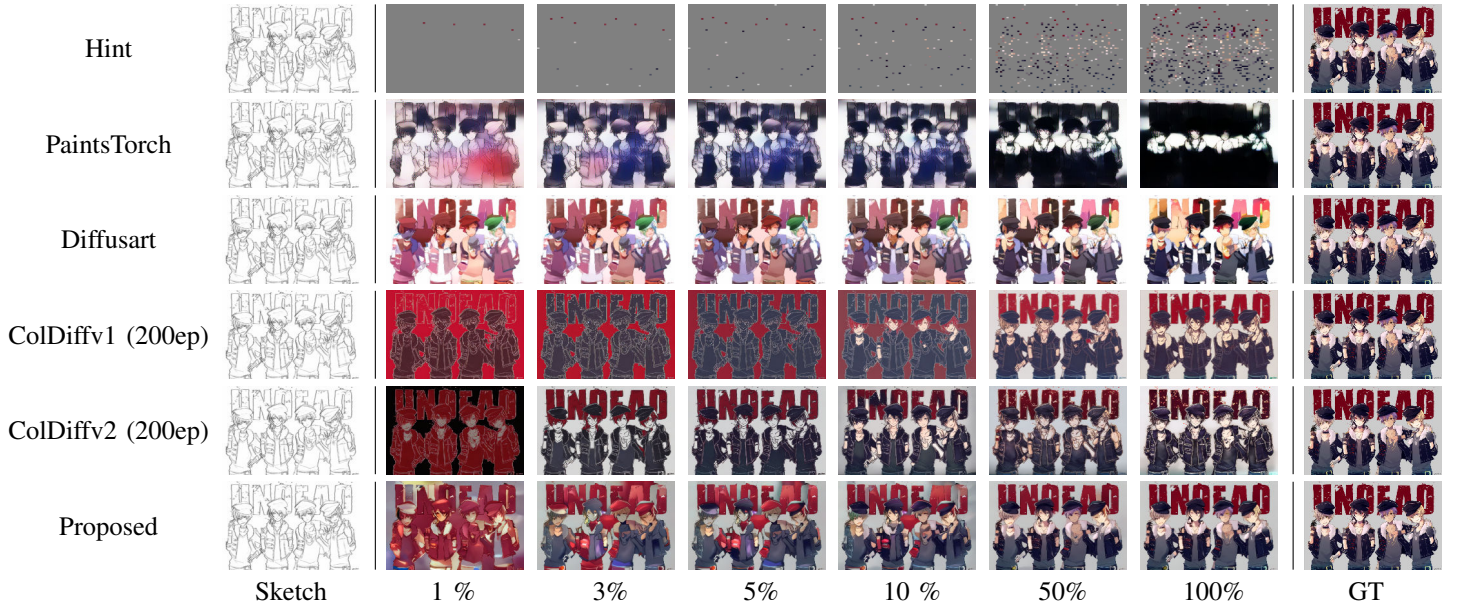


Fig. 11. **Colorization with different hint ratios with deterministic hints by region-based hint sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.



Fig. 12. **Colorization with different hint ratios with deterministic hints by region-based hint sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.



Fig. 13. **Colorization with different hint ratios with deterministic hints by region-based hint sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.



Fig. 14. **Colorization with different hint ratios with deterministic hints by region-based hint sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

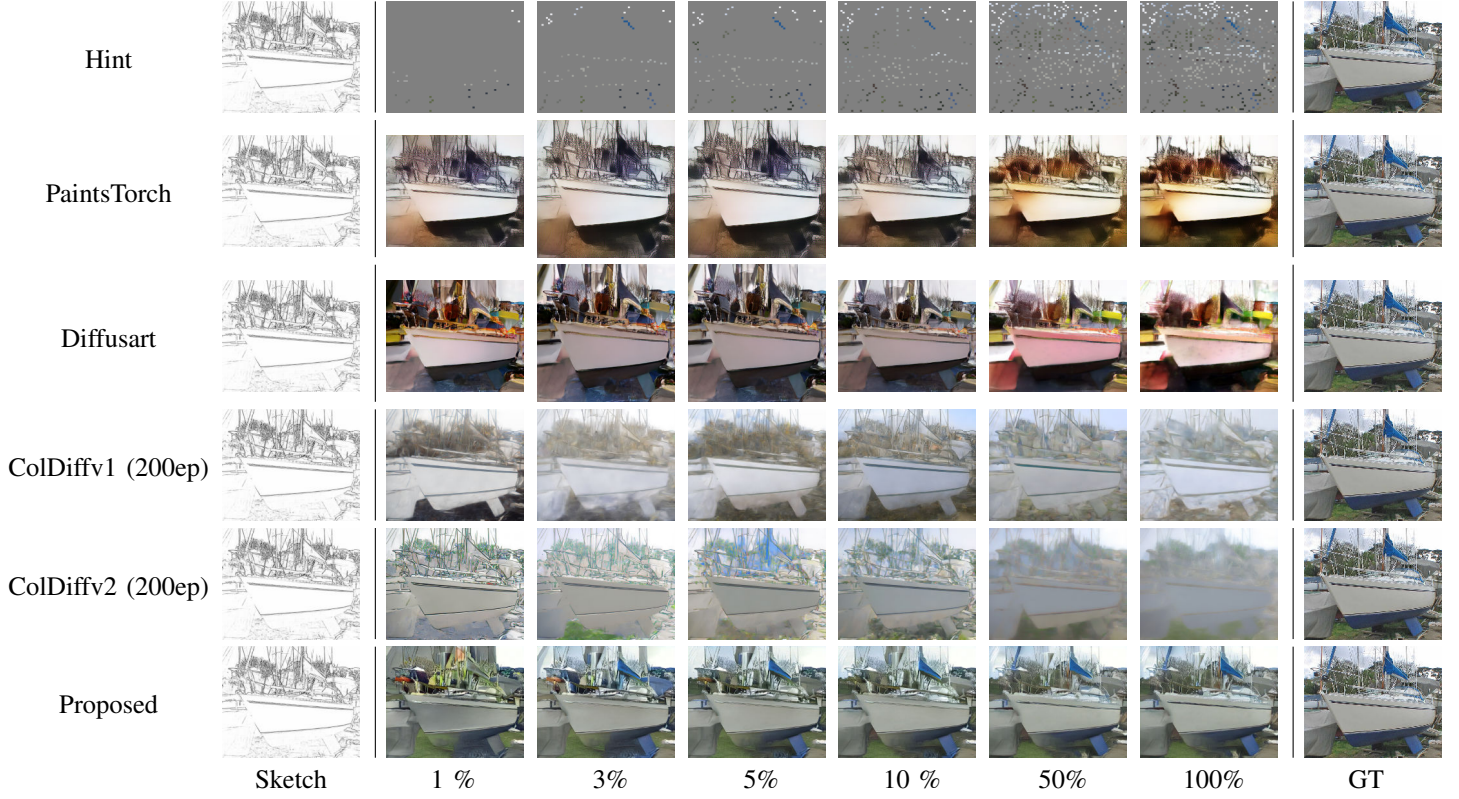


Fig. 15. **Colorization with different hint ratios with deterministic hints by region-based hint sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

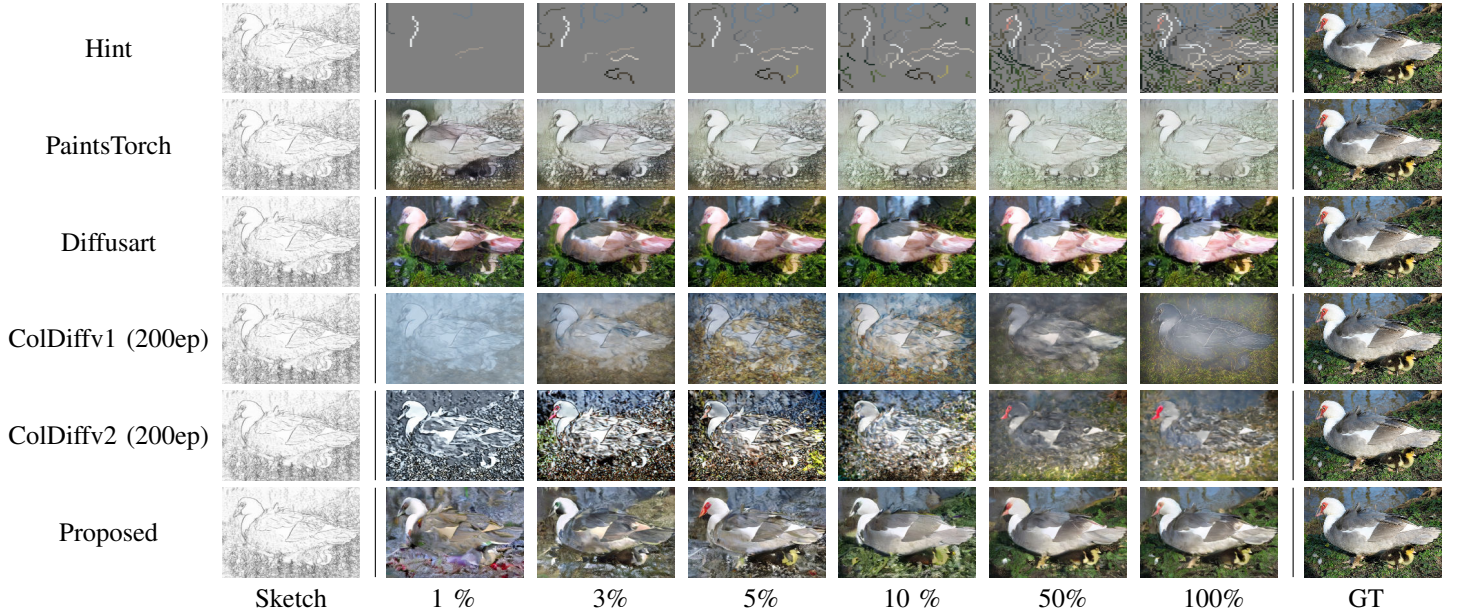


Fig. 16. **Colorization with different hint ratios with deterministic hints by region-based hint sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

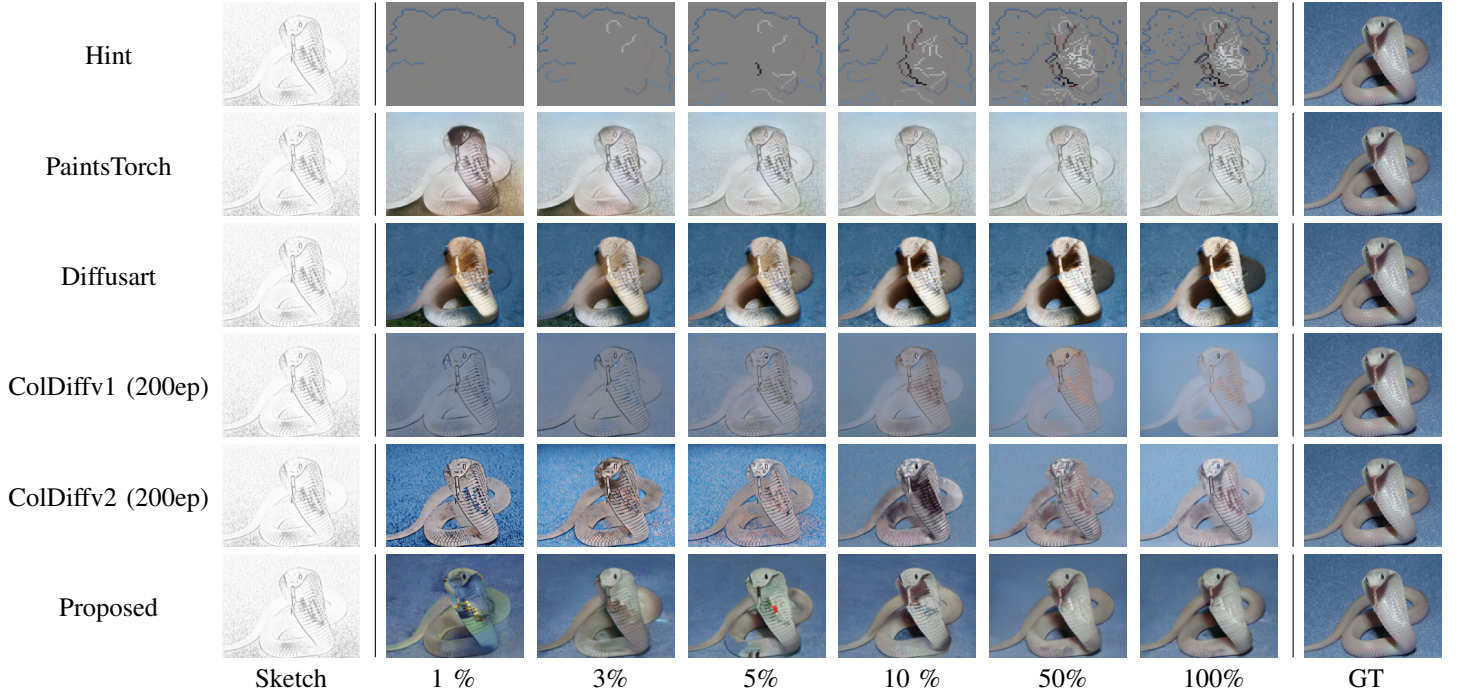


Fig. 17. **Colorization with different hint ratios with deterministic hints by region-based hint sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

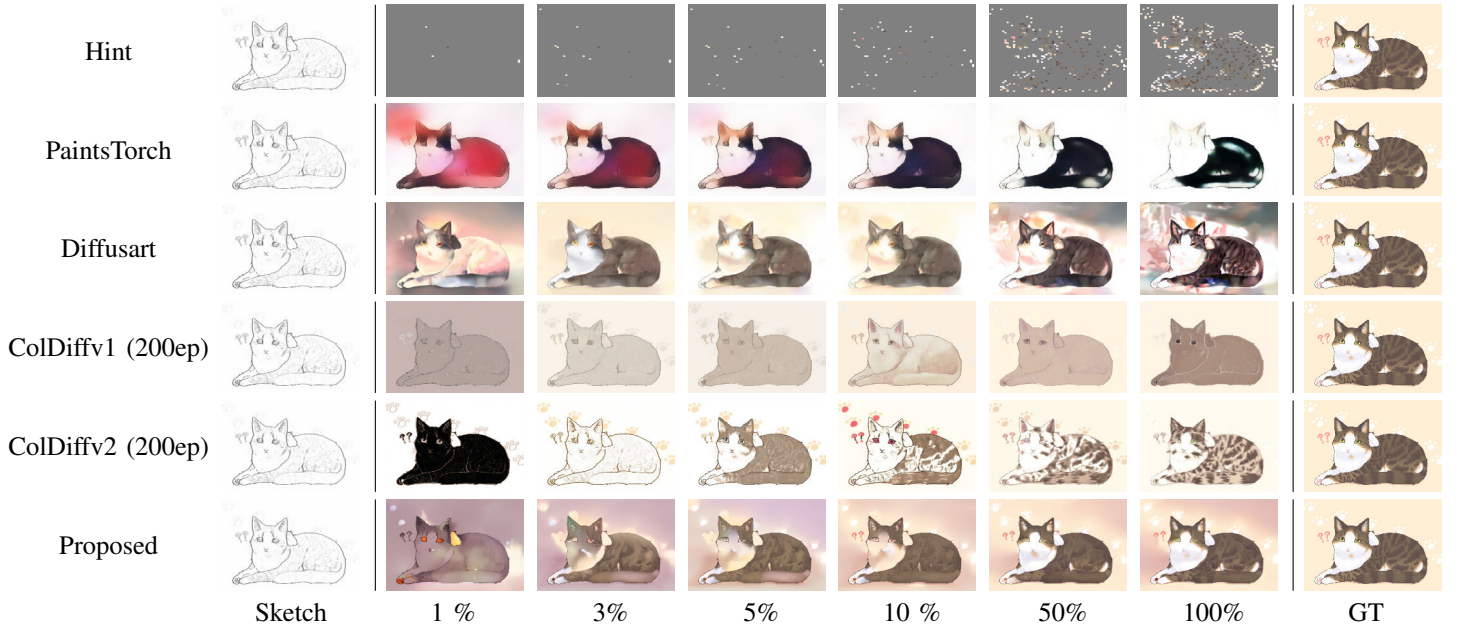


Fig. 18. **Colorization with different hint ratios with deterministic hints by random sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.



Fig. 19. **Colorization with different hint ratios with deterministic hints by random sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

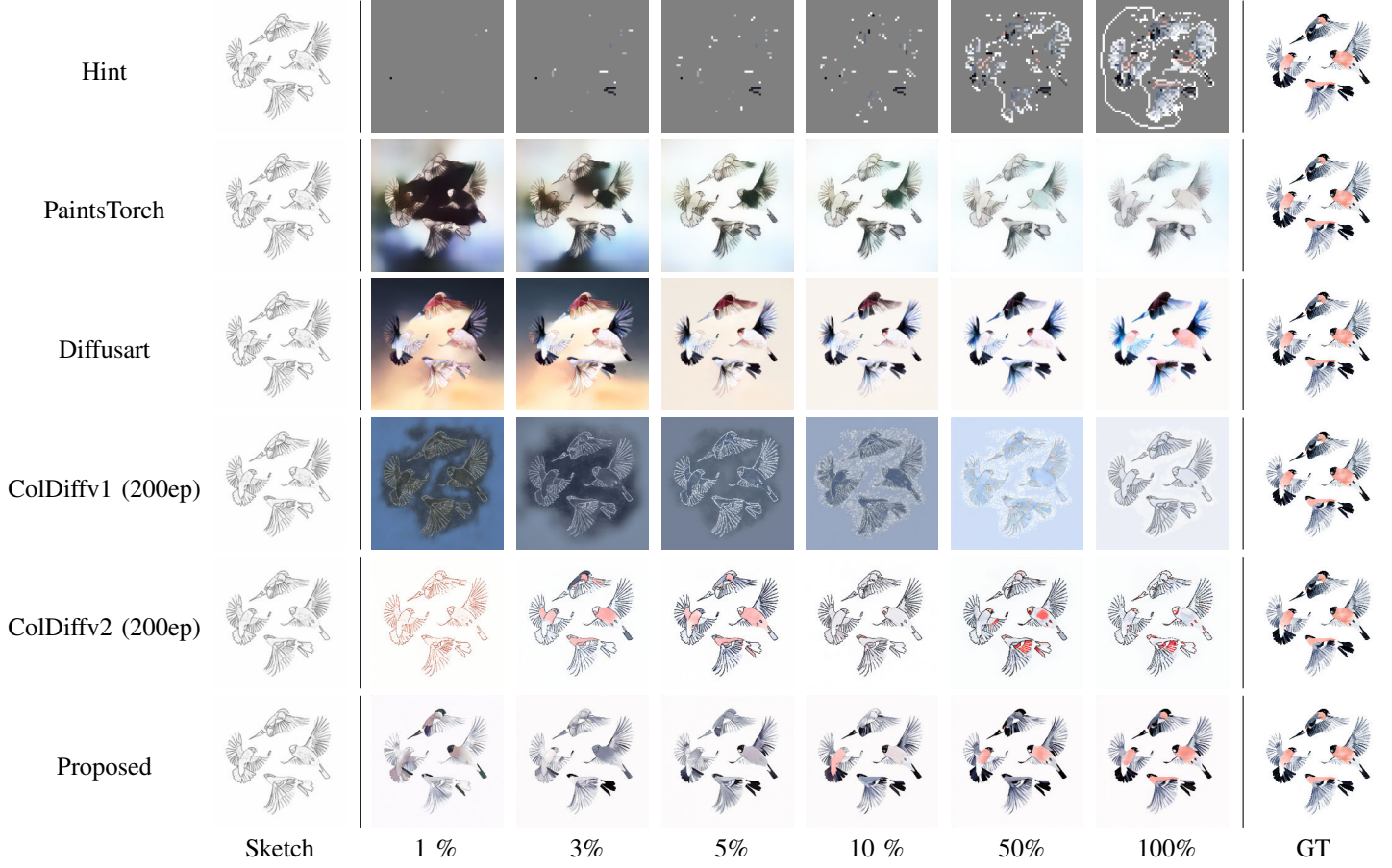


Fig. 20. **Colorization with different hint ratios with deterministic hints by random sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

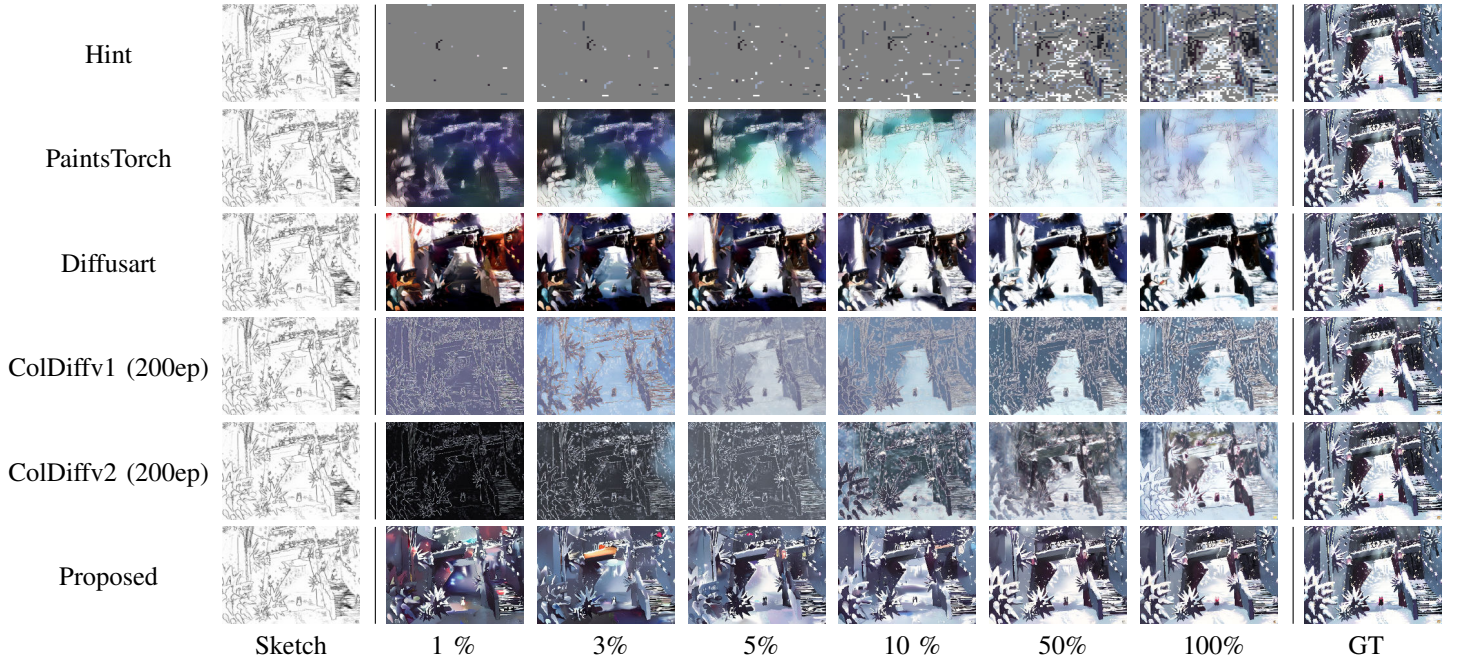


Fig. 21. **Colorization with different hint ratios with deterministic hints by random sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.



Fig. 22. **Colorization with different hint ratios with deterministic hints by random sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

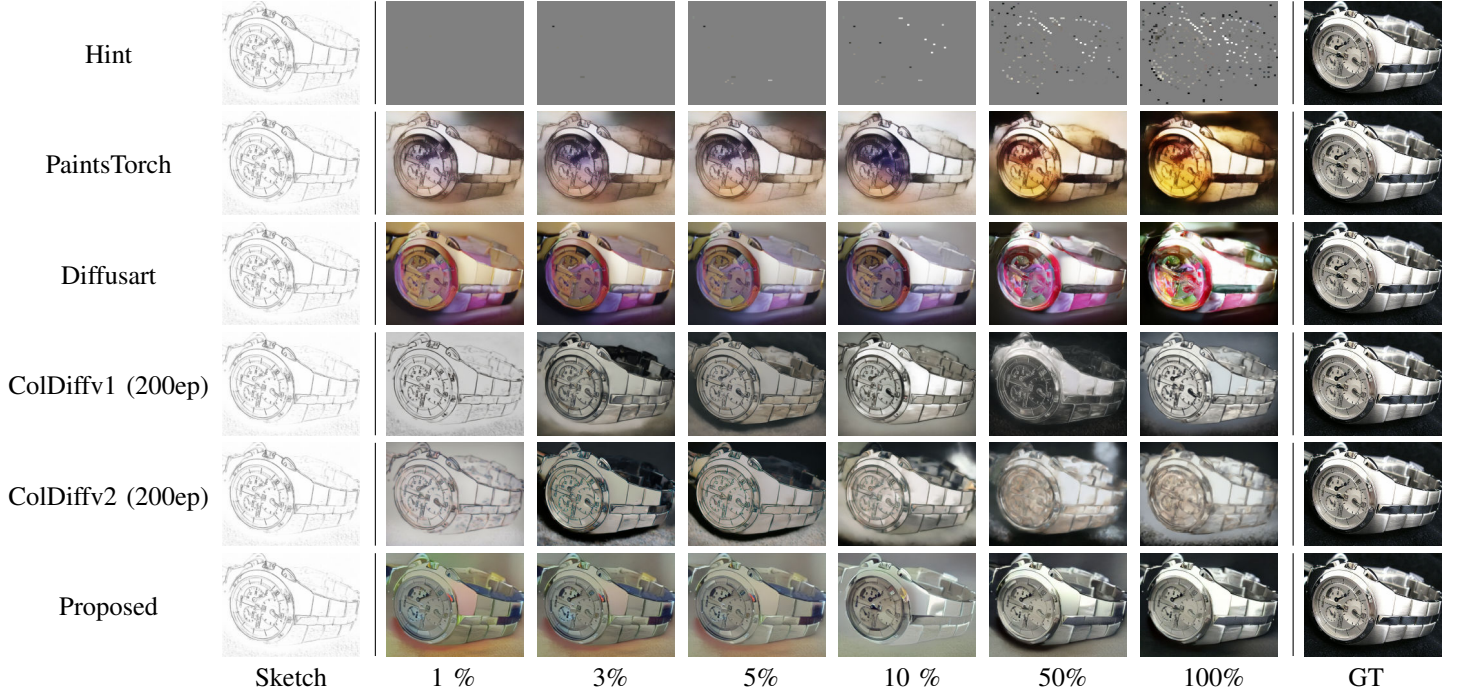


Fig. 23. **Colorization with different hint ratios with deterministic hints by random sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

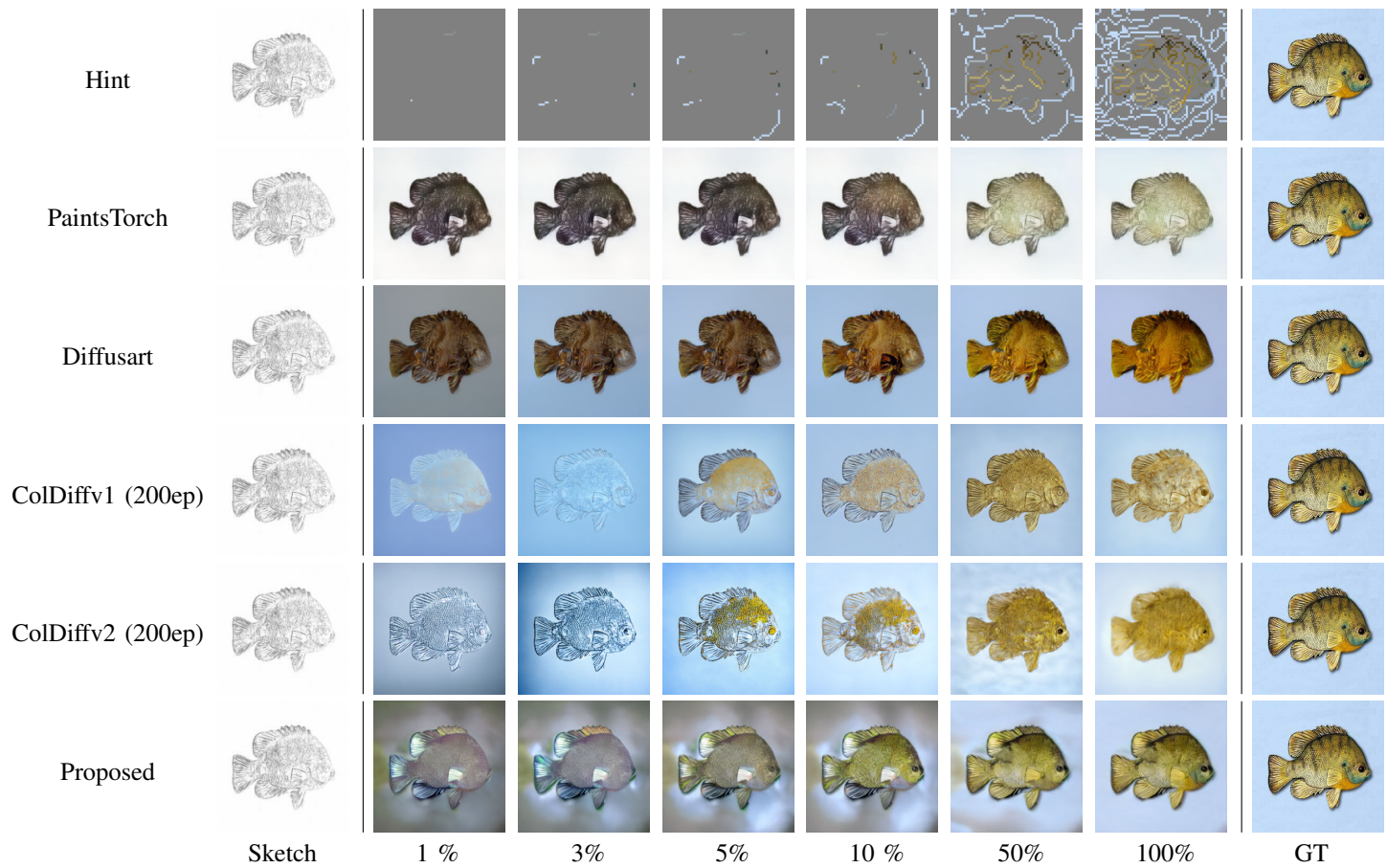


Fig. 24. **Colorization with different hint ratios with deterministic hints by random sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

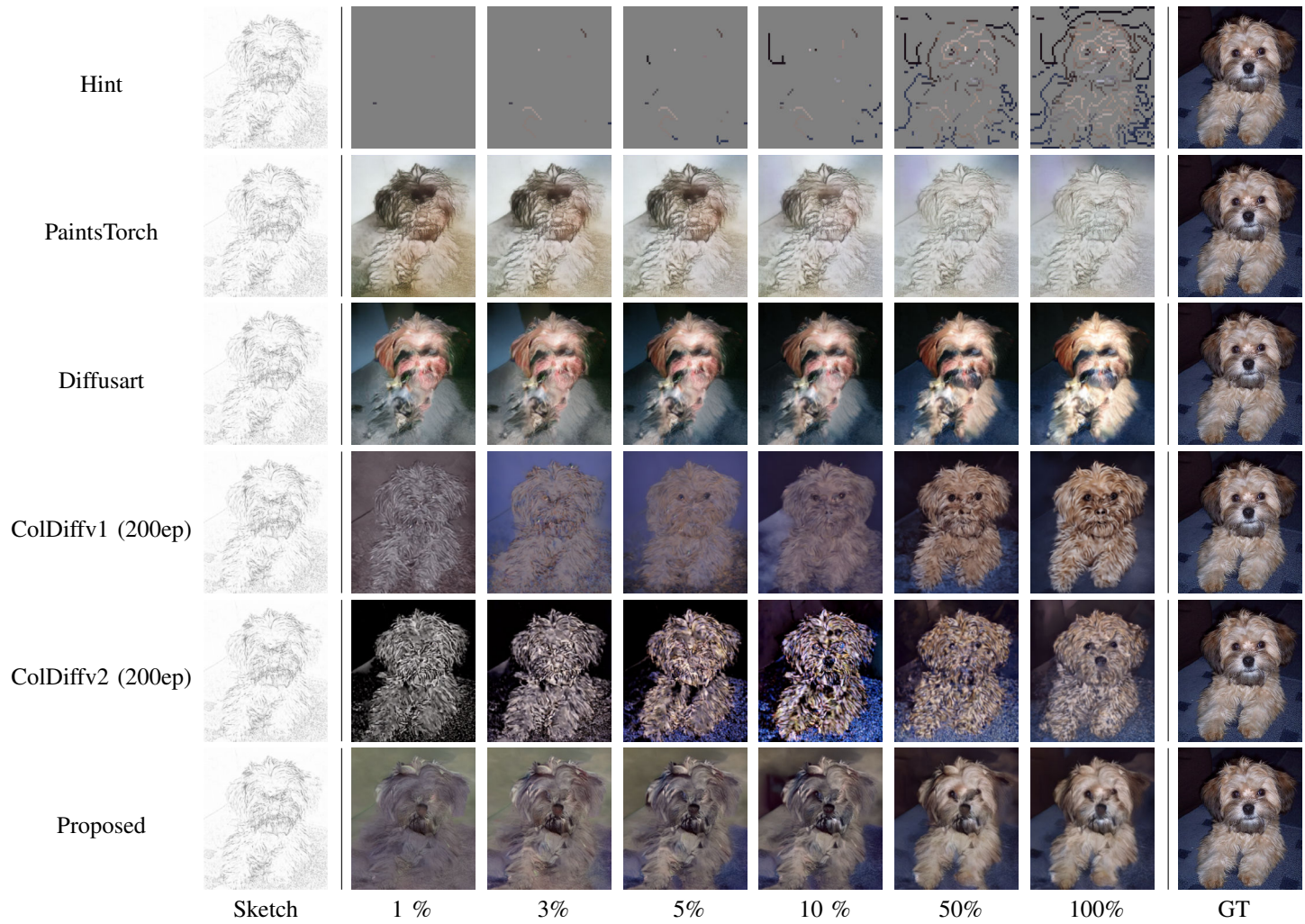


Fig. 25. **Colorization with different hint ratios with deterministic hints by random sampling:** By comparing with several hint ratios α , we found that each model improves the visualization quality in a different way. 200ep: ColorizeDiffusion fine-tuned from the official weights with our 200-epoch protocol.

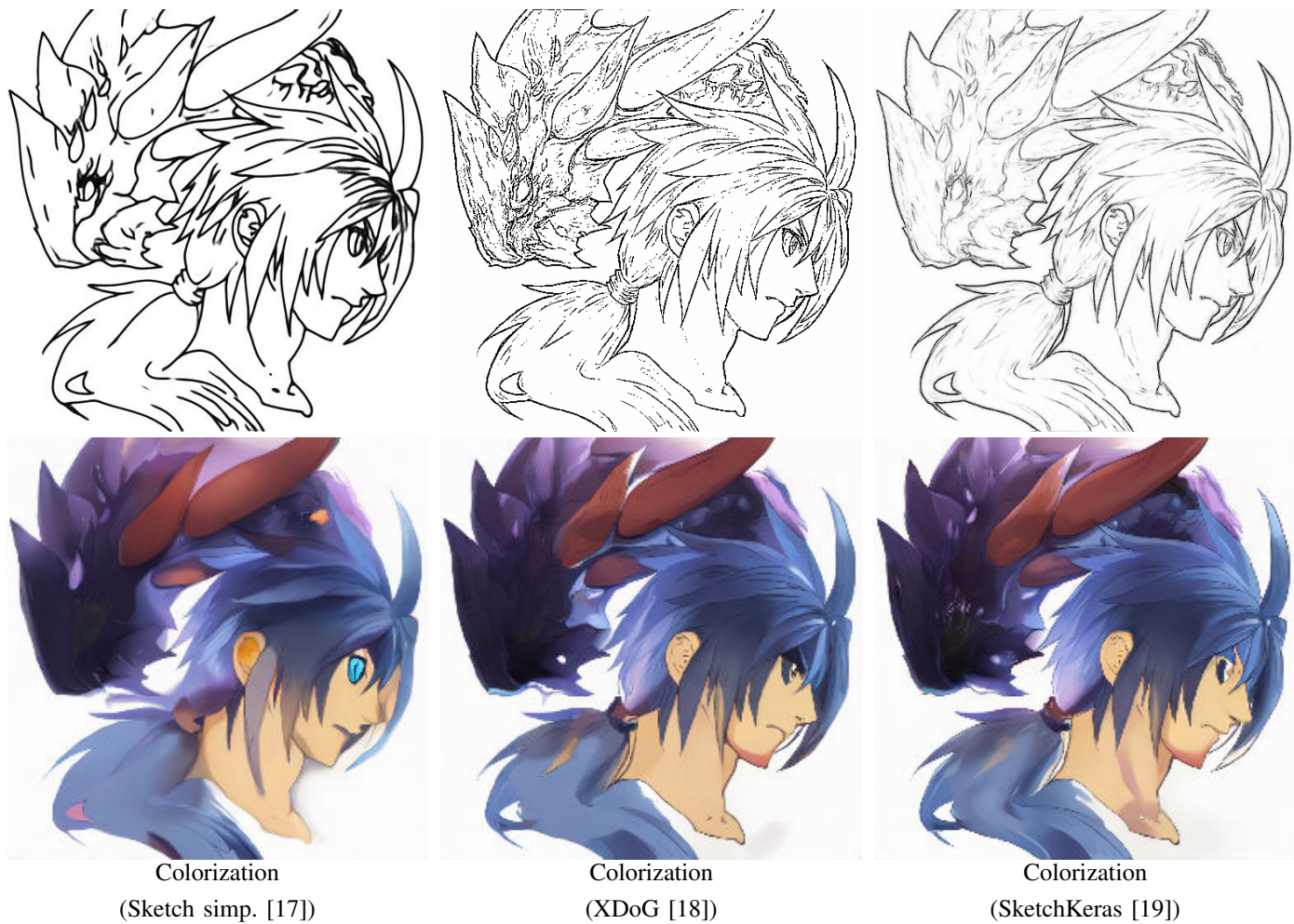


Fig. 26. **Colorization with different sketches with same deterministic hints:** Colorization difference can be confirmed on eyes and hair colors.

TABLE XI
**ADDITIONAL TRAINING COMPARISONS WITH RANDOMLY SAMPLED
 REGION-BASED HINTS ON THE DANBOORU2021 DATASET (HINT-AUC).**
7EP AS IN TABLE X; THE CORRESPONDING 200EP RESULTS ARE
 REPORTED IN TABLE III OF THE MAIN PAPER, AND THE DETFILL ROWS
 ARE REPEATED FROM TABLE III OF THE MAIN PAPER FOR REFERENCE.
 EACH SCORE IS THE MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THE
 THREE SKETCH SOURCES ($n = 3$, DIVISOR $n - 1$).

Hint	Model	MSE ($\downarrow$)	Hint-AUC	
			LPIPS ($\downarrow$)	DreamSim ($\downarrow$)
Dot	ColDiffv1 (7ep) [15]	0.078 $\pm$ 0.011	0.366 $\pm$ 0.035	0.219 $\pm$ 0.021
	ColDiffv2 (7ep) [16]	0.086 $\pm$ 0.010	0.382 $\pm$ 0.026	0.230 $\pm$ 0.017
	DetFill (Ours)	0.025$\pm$0.005	0.220$\pm$0.031	0.139$\pm$0.020
Scribble	ColDiffv1 (7ep) [15]	0.069 $\pm$ 0.010	0.344 $\pm$ 0.033	0.198 $\pm$ 0.018
	ColDiffv2 (7ep) [16]	0.119 $\pm$ 0.015	0.390 $\pm$ 0.032	0.235 $\pm$ 0.019
	DetFill (Ours)	0.021$\pm$0.007	0.194$\pm$0.045	0.091$\pm$0.020

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